



ELEC 5690 - Advanced Topics in Artificial Intelligence for Medical Image Analysis

Lecture 01a: Course Introduction

Sep 1, 2026

Who's who

Instructor

Xiaomeng LI
Associate Professor, ECE



Office: Rm 2524
eexmli@ust.hk

Teaching Assistant

Xinrui ZHOU
PhD student, ECE



Office: Rm 3117
xzhoucv@connect.ust.hk
Medical Image Analysis, Point Cloud
Registration, Generative Model

Office Hour: Tue, 3:00 - 4:00 pm

Why learn this course?

AI in healthcare: a rapidly exploding field



Hospitals Roll Out AI Systems to Keep Patients From Dying of Sepsis

Septic shock kills 50 percent of people who are affected—Sepsis Watch could save their lives

By Eliza Strickland

In hospitals, doctors and nurses keep vigilant watch over patients'



Google, Verily using AI to screen for diabetic retinopathy in India

The machine learning algorithm can also help with screening for diabetic macular edema, a boon for patients in a country where physicians are in short supply.

By Mike Miliard | February 26, 2019 | 03:17 PM



(Photo: Google)

- Introduce you to the exciting field of AI in healthcare, particularly medical image analysis.
- Learn basics of deep neural networks and medical images
- Learn how AI works for healthcare and much more...

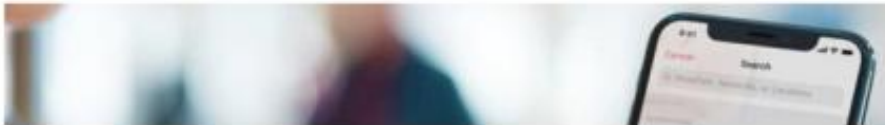
Why learn this course?

AI in healthcare: a rapidly exploding field

Apple's future healthcare market moves will rely heavily on AI analysis

By Malcolm Owen
Monday, September 16, 2019, 09:03 am PT (12:03 pm ET)

Apple's moves in the healthcare market could involve the tracking of user data for further analysis by artificial intelligence and billing model based on cost-savings, with analysts pointing out areas of the consumer health industry Apple could easily advance by building upon its already-released technology and services.



Google to Store and Analyze Millions of Health Records

The tech company's deal with Ascension is part of a push to use artificial intelligence to aid health services.



Amazon Makes Healthcare Buy As Its Plans Start to Take Shape

The e-commerce giant just bought a telemedicine technology startup to help with its healthcare-related efforts.



Donna Fuscaldo (TMFdonnabail)
New York, 2019/09/16/11:03AM

In what is its first healthcare-related acquisition since spending \$753 million in June 2018 to acquire PillPack, **Amazon.com** ([NASDAQ:AMZN](#)) inked a deal to buy Health Navigator, a start-up that provides digital triage tools and symptom lookup. The value of the deal was not disclosed.

Amazon intends to offer Health Navigator services to its employees, shedding further light on where the e-commerce giant is heading in the healthcare market.



MICROSOFT | SCIENCE | TECH

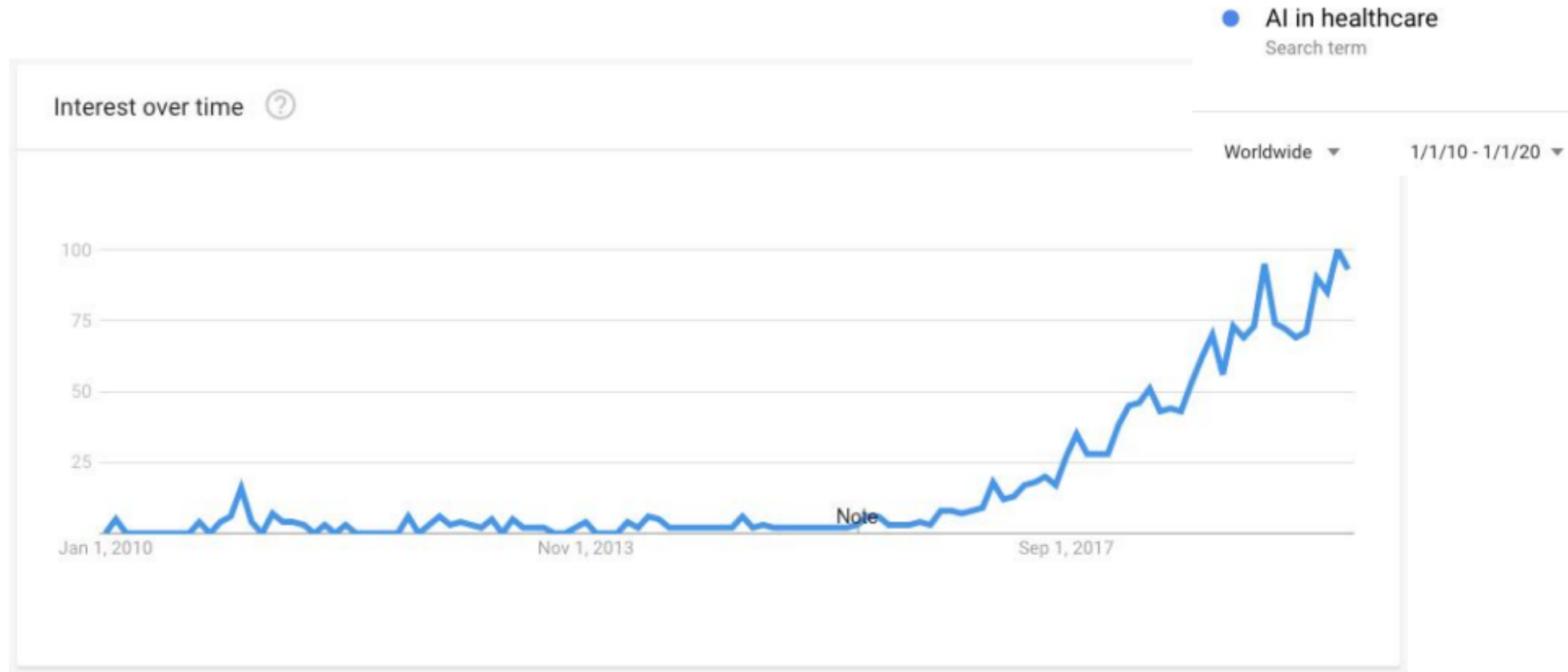
Microsoft Healthcare is a new effort to push doctors to the cloud

Microsoft wants to be a big part of the cloud and AI healthcare race

By Tom Warren | [@tomwarren](#) | Jun 27, 2018, 6:50am EDT

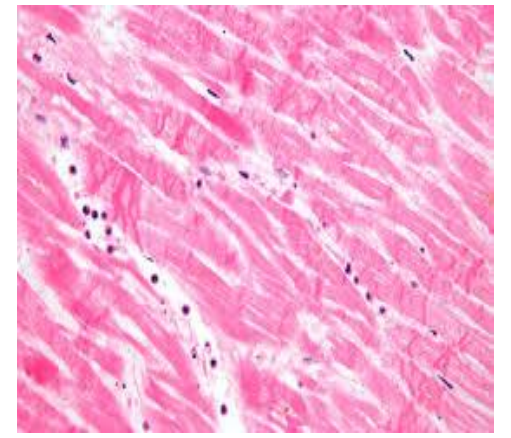
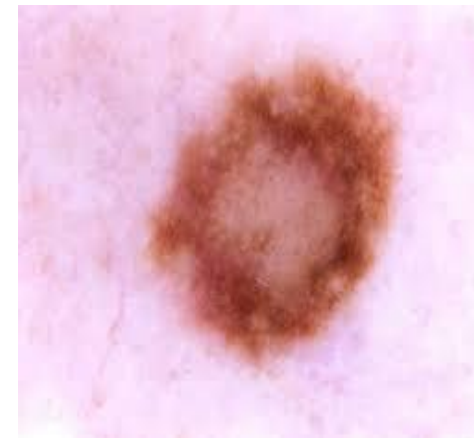
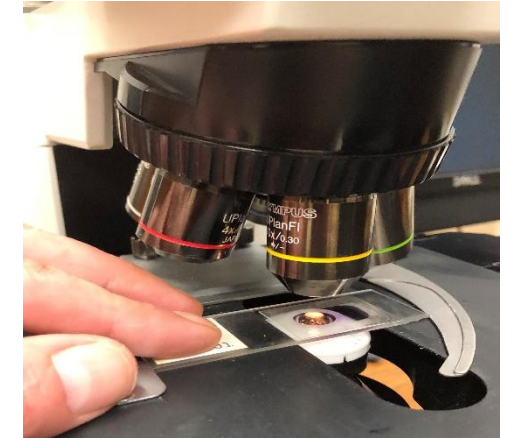
Why learn this course?

AI in healthcare: a rapidly exploding field



Why learn this course?

Medical images make up around 90% of the data in healthcare.



X-rays

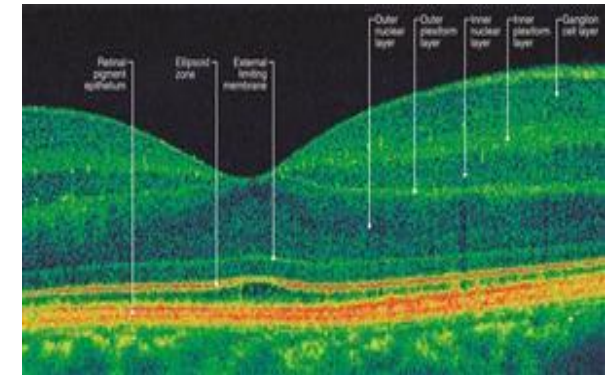
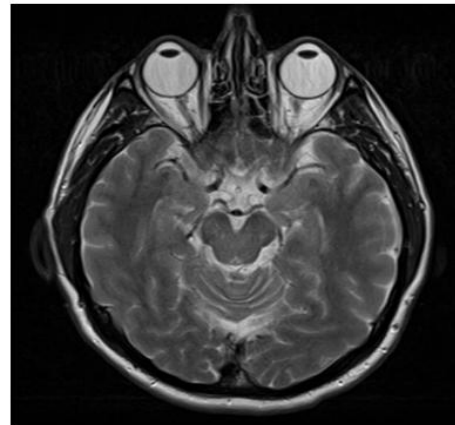
Fundus Photography

Dermoscopy

Histopathology

Why learn this course?

Medical images make up around 90% of the data in healthcare.



Computed tomography (CT)

Magnetic resonance imaging (MRI)

Optical coherence tomography (OCT)

This course

Artificial intelligence, machine learning, deep learning

Artificial intelligence

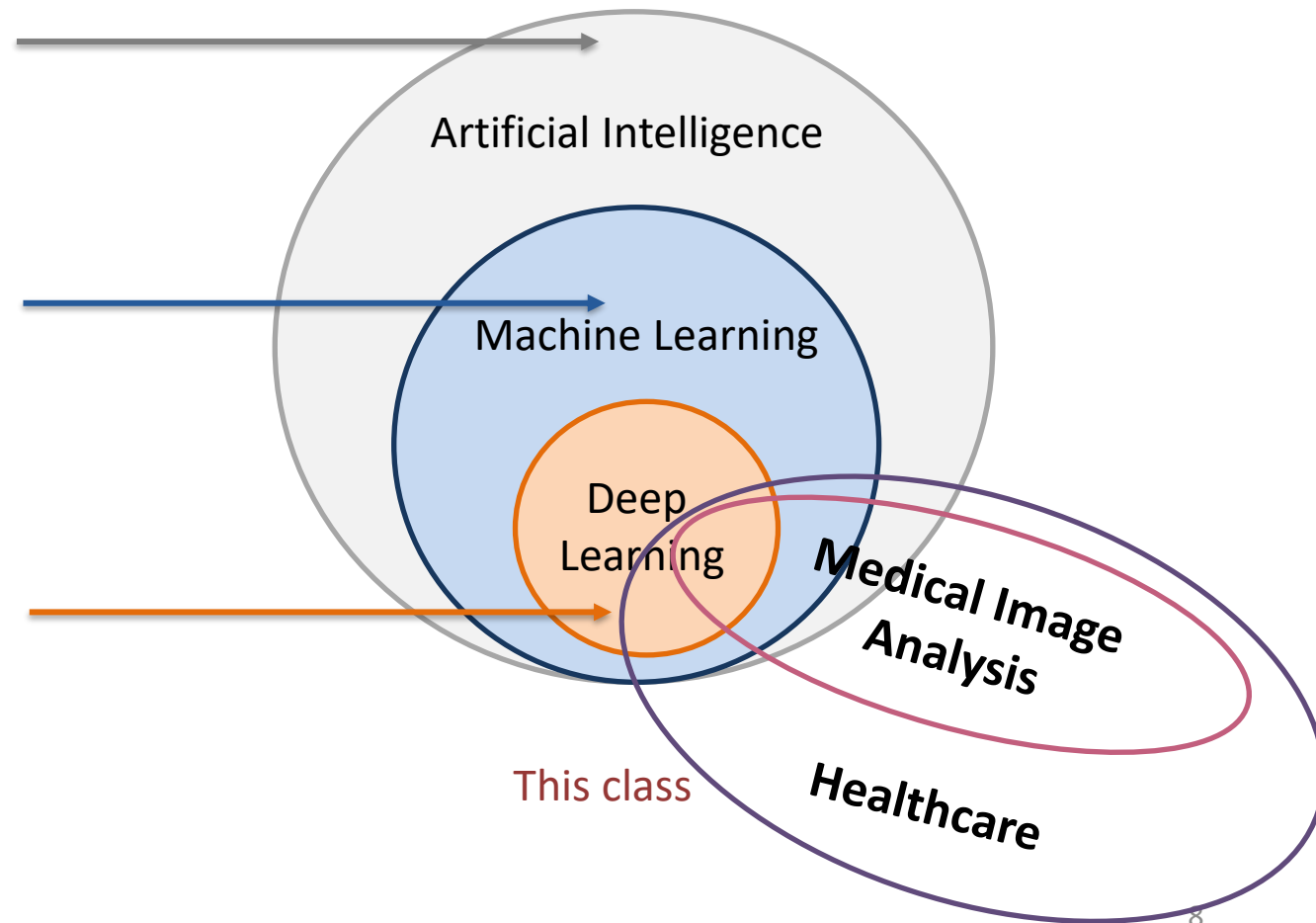
Any technique which enables computers to mimic human behavior.

Machine Learning

Subset of AI techniques which use **statistical methods** to enable machines to improve with experiences.

Deep Learning

Subset of ML which make the computation of **multi-layer neural networks** feasible.

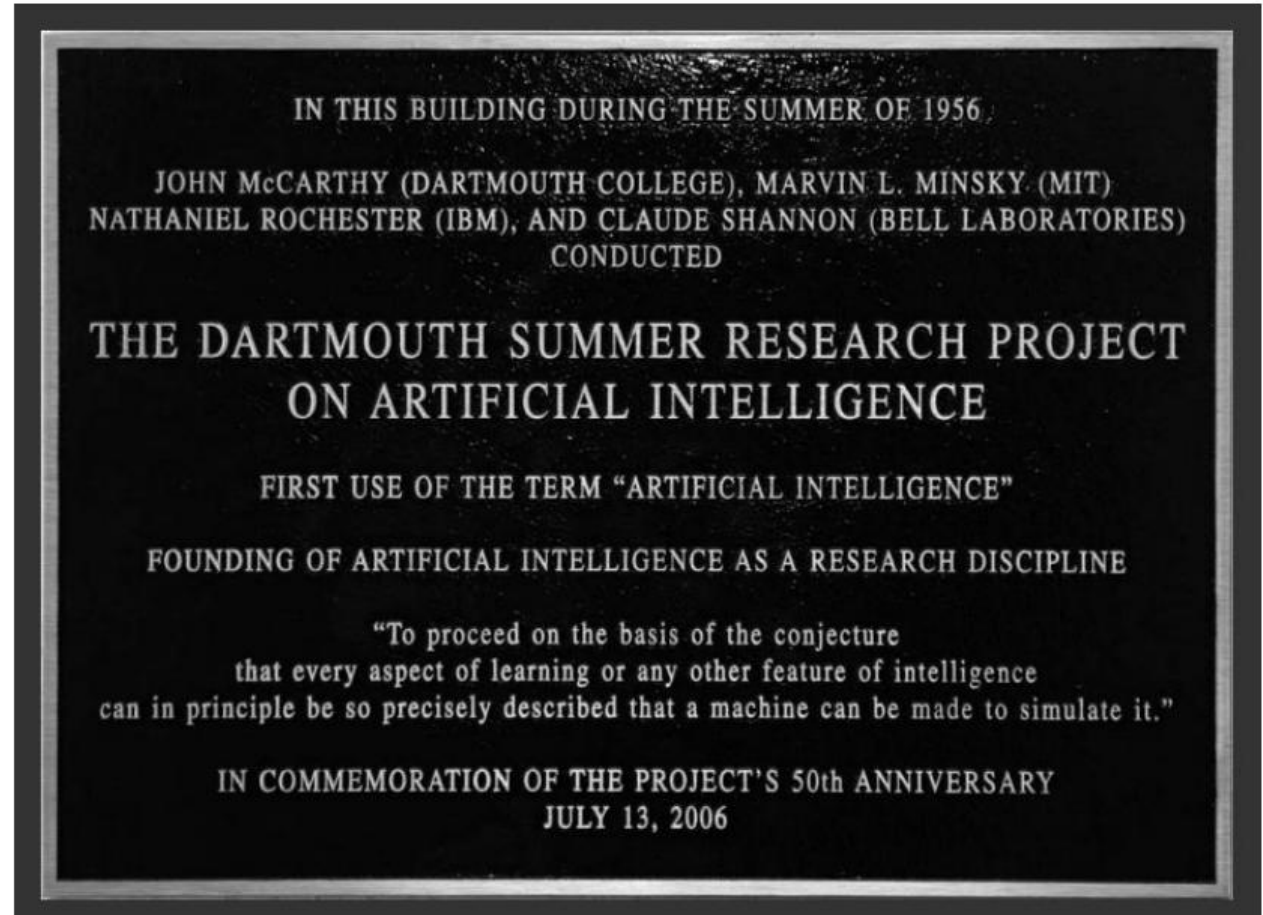


A brief history of AI

- 1956: Birth of AI as a modern research discipline



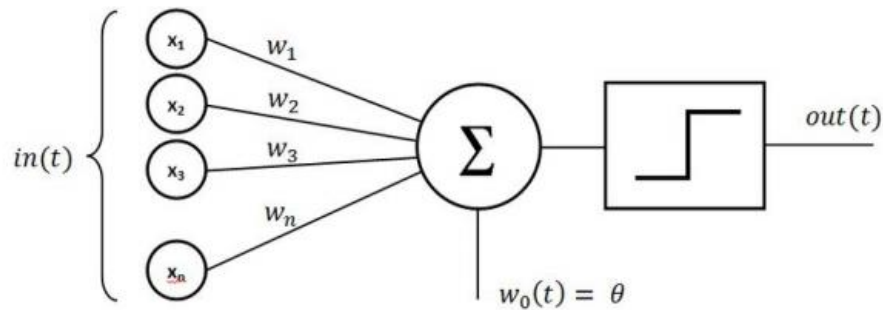
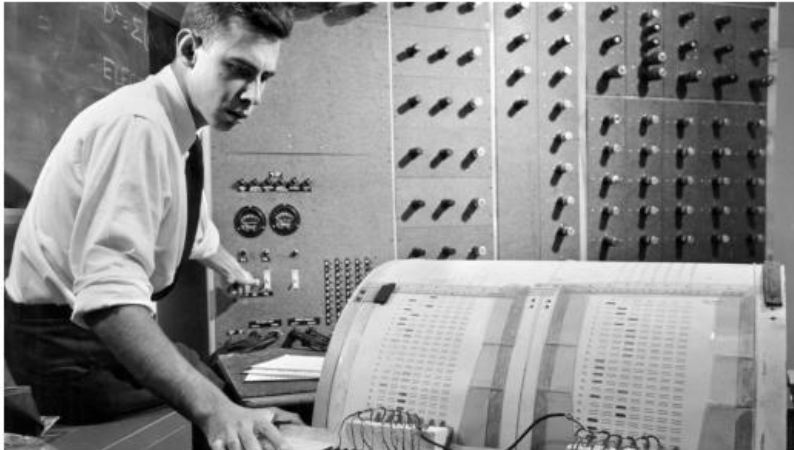
John McCarthy



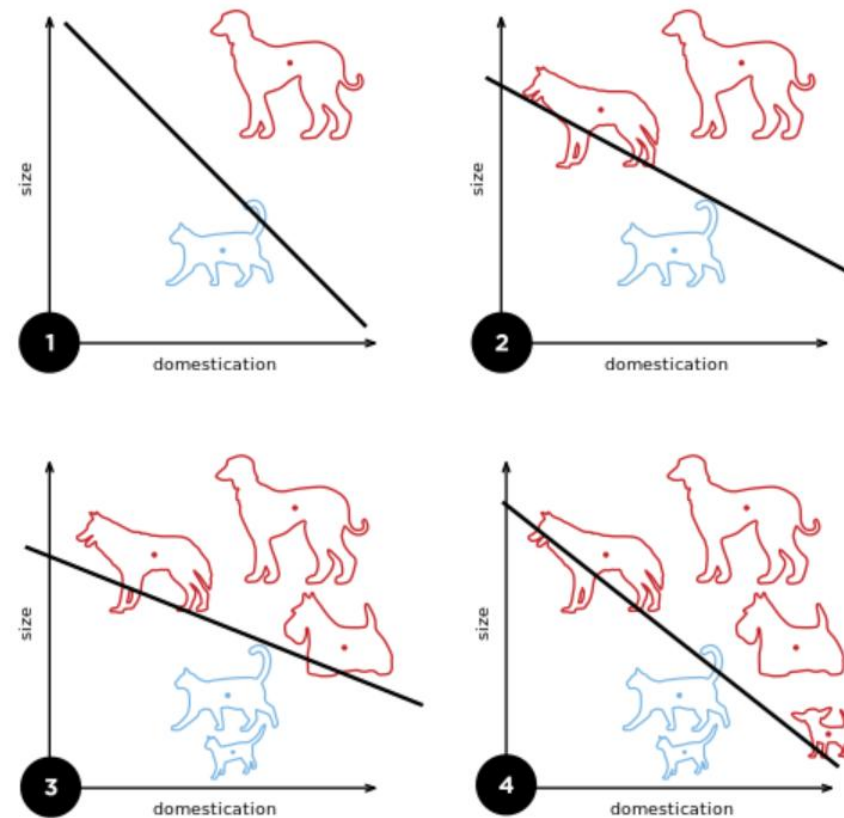
[picture from google]

A brief history of AI

- 1950-1960s: Early progress of AI

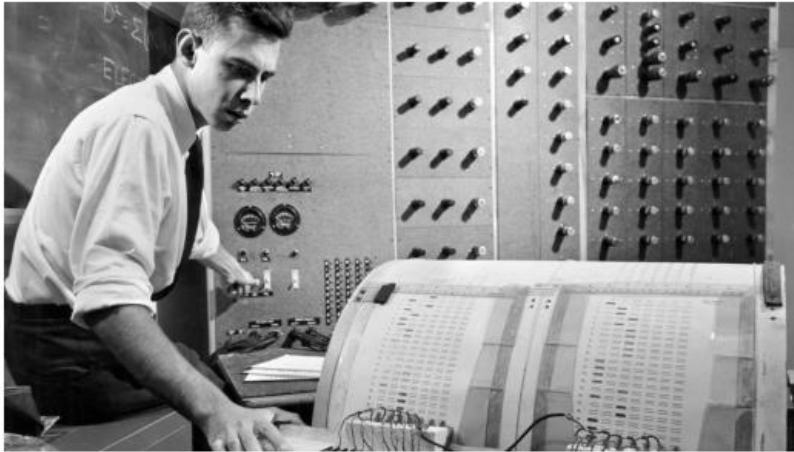


Perceptron model: Rosenblatt, 1958

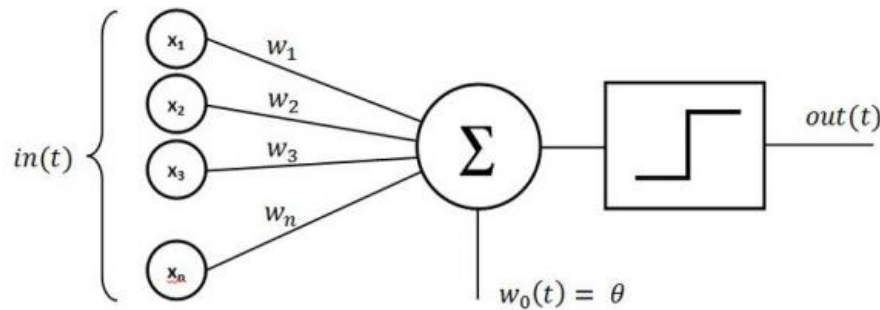


A brief history of AI

- 1950-1960s: Early progress of AI



- Research came to a sudden halt after publication of a book “Perceptrons” in 1969 by Minsky and Papert.
- It suggested that there were severe limitations of perceptron.
- No research at all was done for nearly 10 years.
- A new generation of research revived the field and thereafter found it was a useful part of artificial intelligence.



Perceptron model: Rosenblatt, 1958

A brief history of AI

- 1950-1960s: Early progress of AI

```
Welcome to

      EEEEE LL      IIII ZZZZZZZZ  AAAAA
      EE     LL      II      ZZ  AA  AA
      EEEEE LL      II      ZZZ  AAAAAA
      EE     LL      II      ZZ  AA  AA
      EEEEE LLLLLL IIII ZZZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:   █
```

Capabilities of early projects soon reached limitations they could not surpass.

Enthusiasm for AI dwindled in the 70s.

ELIZA chatbot: Weizenbaum, 1966

A brief history of AI

- 1980s: Resurgence

```
#DEFINE MOLFORM C 12 H 14 O
MOLECULAR FORMULA DEFINED
```

```
#DEFINE SUBSTRUCTURE Z
```

[Z is the structure required by constraints C2 and C3.]

■ ■ ■

```
CONSTRAINT: SUBSTRUCTURE CH0 EXACTLY 2
```

[C7: we must end up with exactly two quaternary carbons.]

```
CONSTRAINT: RING 3 NONE [C9]
```

```
CONSTRAINT: RING 4 NONE [C9]
```

```
CONSTRAINT:
```

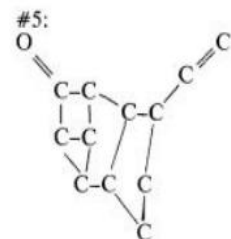
```
#####
```

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#####
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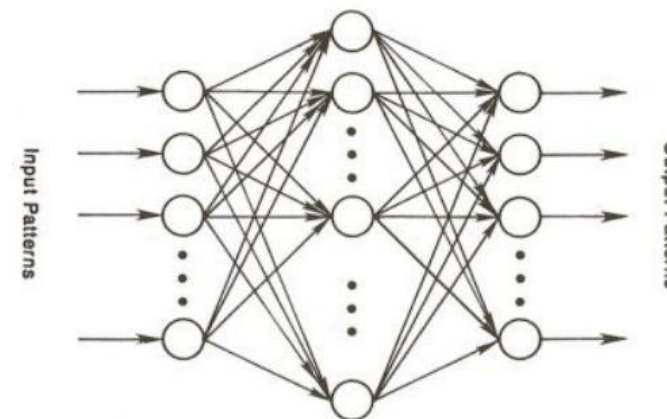
```
47 STRUCTURES WERE OBTAINED
```

```
#DRAW ATNAMED (5 6)
```

[Comment: The following is a selection of final structures 5, 6.]



Expert systems, 1970s and 80s.
Feigenbaum, etc.



To get the correct generalization of the delta rule, we must set

$$\Delta_p w_{ji} \propto - \frac{\partial E_p}{\partial w_{ji}},$$

where E is the same sum-squared error function defined earlier. As in the standard delta rule it is again useful to see this derivative as resulting from the product of two parts: one part reflecting the change in error as a function of the change in the net input to the unit and one part representing the effect of changing a particular weight on the net input. Thus we can write

$$\frac{\partial E_p}{\partial w_{ji}} = \frac{\partial E_p}{\partial net_{pj}} \frac{\partial net_{pj}}{\partial w_{ji}}. \quad (9)$$

By Equation 7 we see that the second factor is

$$\frac{\partial net_{pj}}{\partial w_{ji}} = \frac{\partial}{\partial w_{ji}} \sum_k w_{jk} o_{pk} = o_{pi}. \quad (10)$$

Backpropagation. Rumelhart, 1986.

Breakthroughs in deep learning

- 1990s: First appearances of modern neural networks

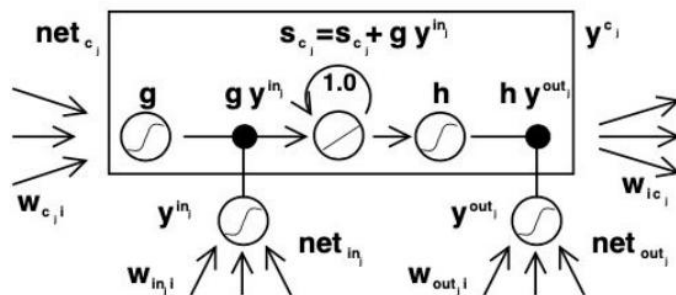
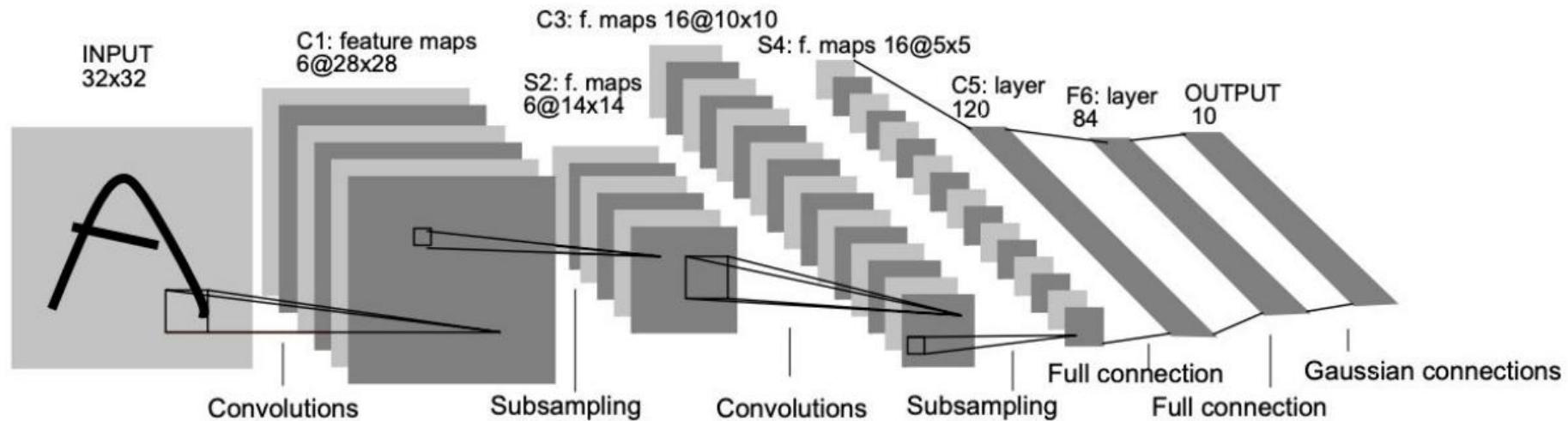


Figure 1: Architecture of memory cell c_i (the box) and its gate units in_i, out_i . The self-recurrent connection (with weight 1.0) indicates feedback with a delay of 1 time step. It builds the basis of the "constant error carousel" CEC. The gate units open and close access to CEC. See text and appendix A.1 for details.

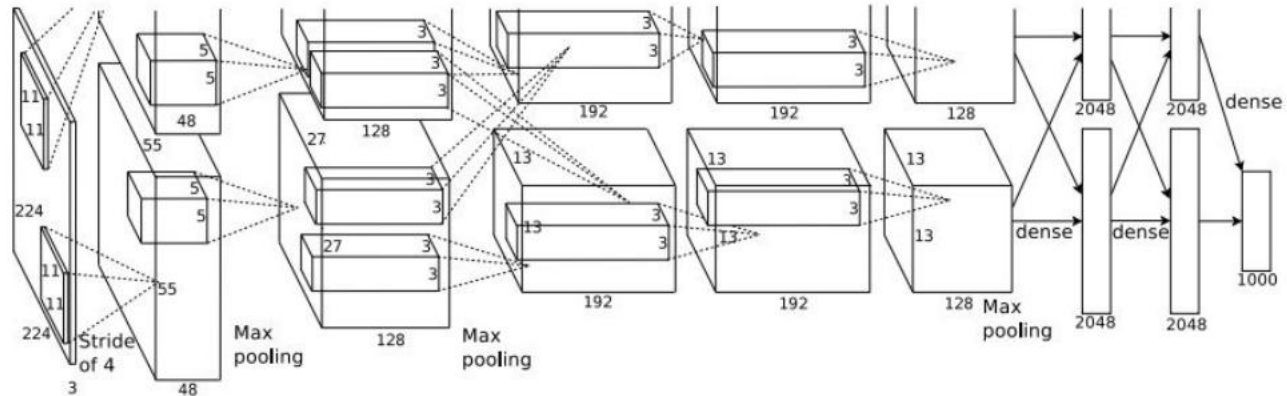
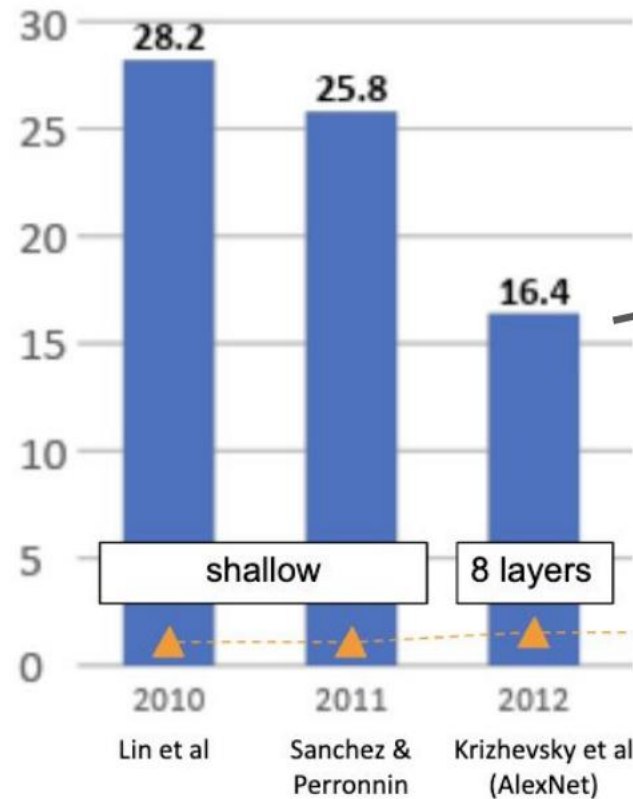
Schmidhuber, 1997.

But limitations reached again...
-> AI came to winter of the 1990s and 2000s

A type of recurrent neural network called the long short-term memory (LSTM)

Breakthroughs in deep learning

- 2012: deep learning breakthrough



Krizhevsky et al. 2012. 8-layer “AlexNet”.

ImageNet Visual Recognition Challenge results.

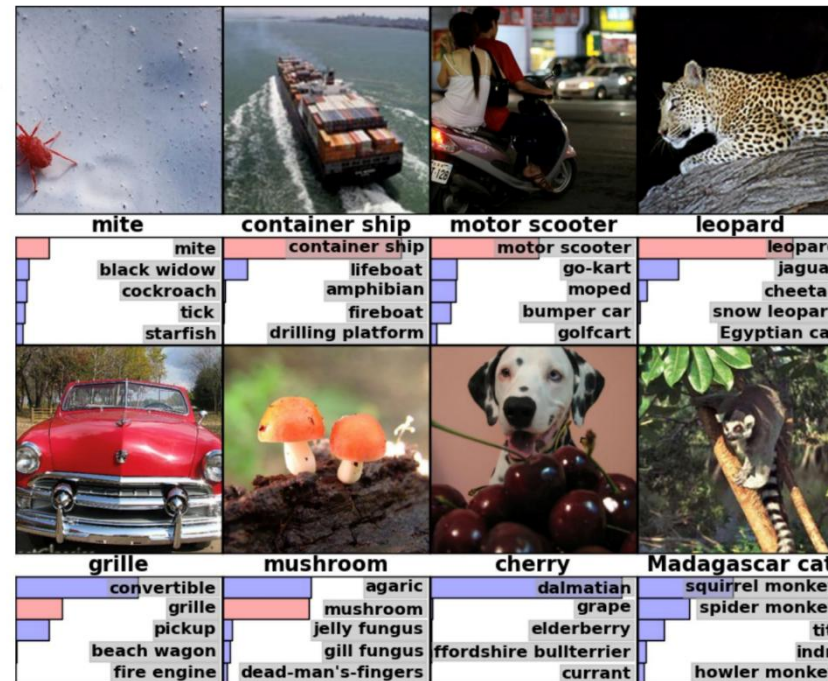
Breakthroughs in deep learning

- 2012: deep learning breakthrough

ImageNet Challenge

IMAGENET

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

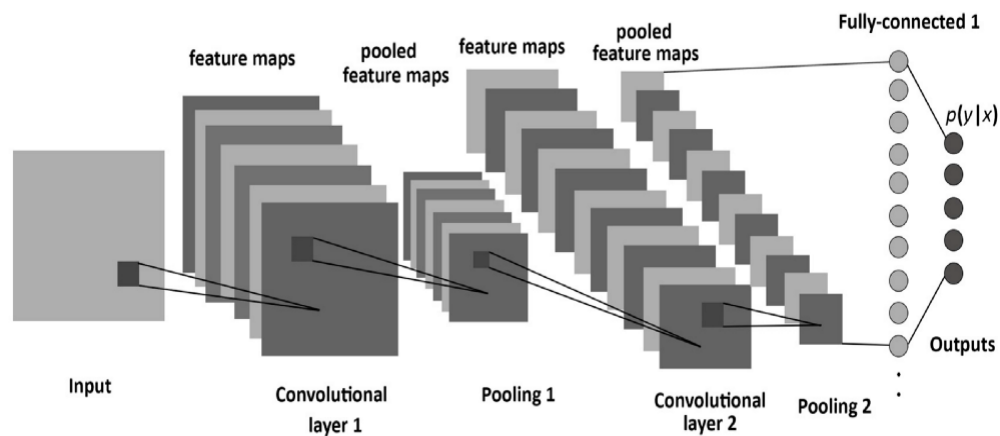


Fei-Fei Li, Stanford University

Breakthroughs in deep learning

- Why success?

Neural networks



Powerful Computation

GPUS MAKE DEEP LEARNING ACCESSIBLE

Deep learning with COTS HPC systems
A. Coates, B. Huval, T. Wang, D. Wu, A. Ng, B. Catanzaro
ICML 2013

“Now You Can Build Google’s \$1M Artificial Brain on the Cheap”

WIRED

GOOGLE DATACENTER

1,000 CPU Servers
2,000 CPUs + 16,000 cores
600 kWatts
\$5,000,000

STANFORD AI LAB

3 GPU-Accelerated Servers
12 GPUs + 18,432 cores
4 kWatts
\$33,000

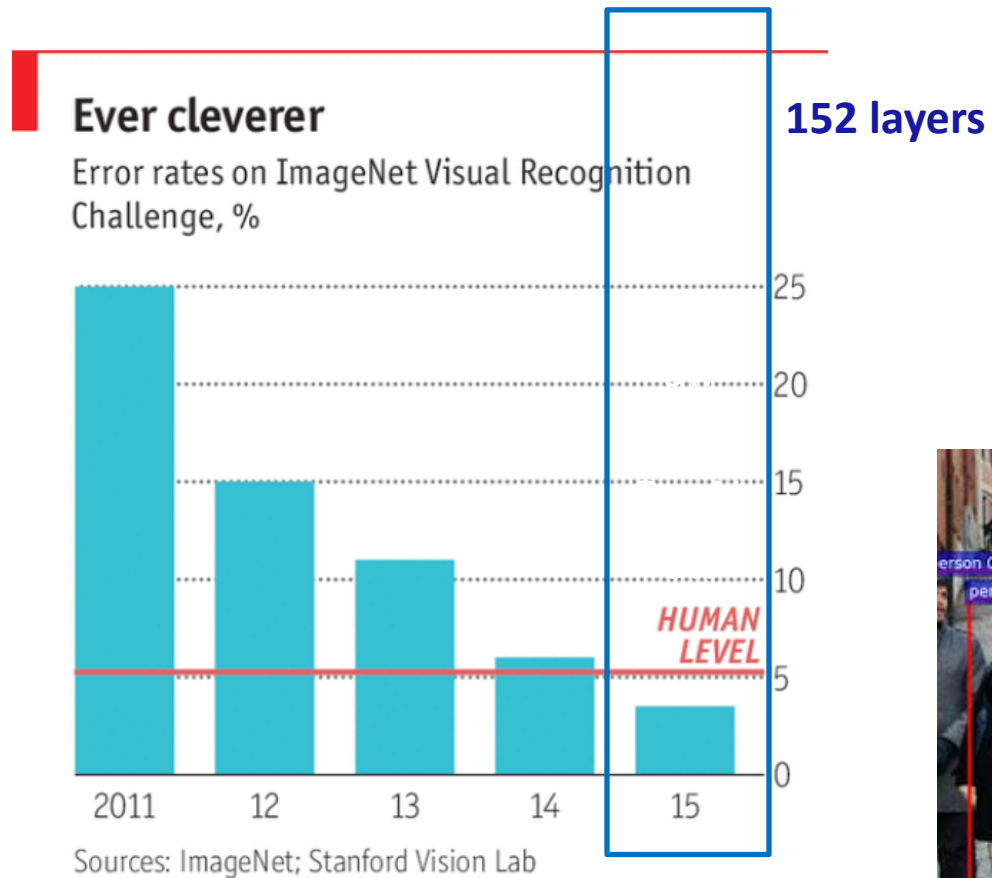
21 NVIDIA

Data

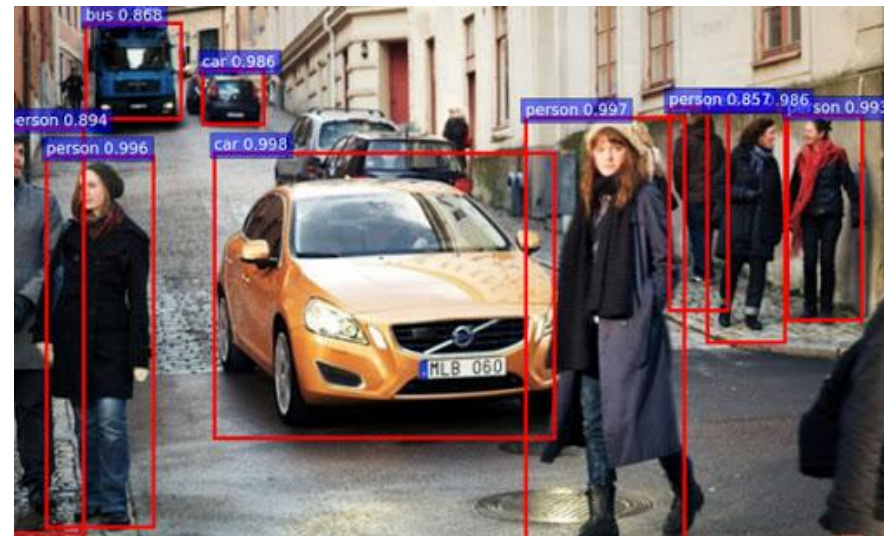
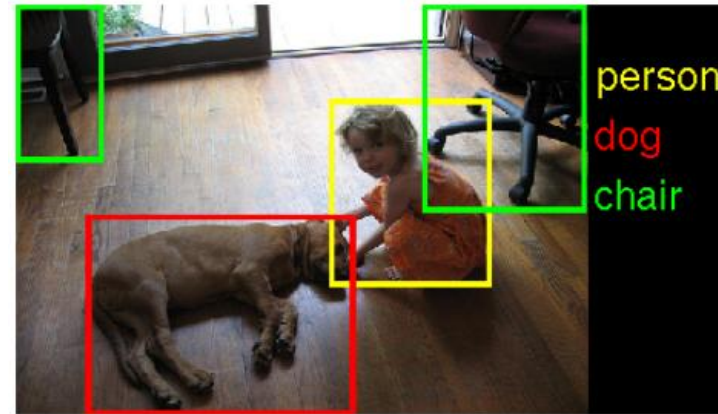


Breakthroughs in deep learning

- 2015: Very deep convnets and challenging vision tasks

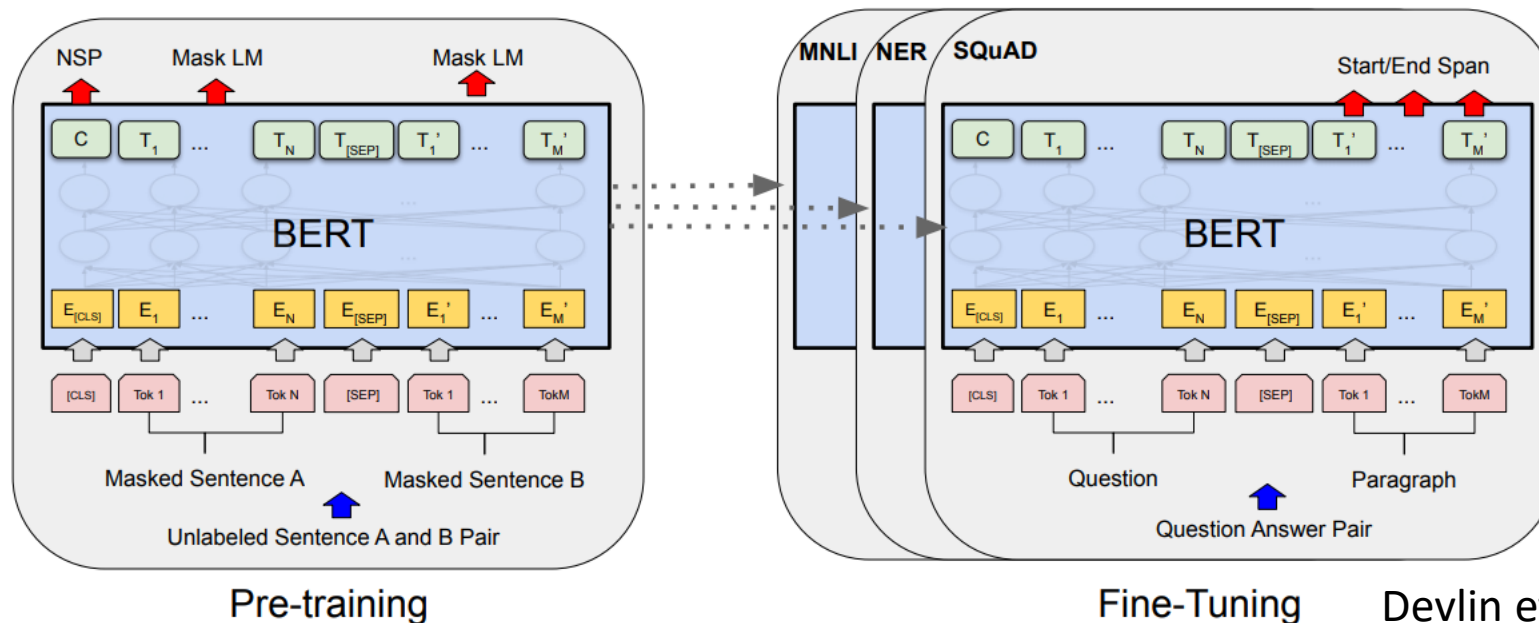


He et al. 2015. ResNet.



Breakthroughs in deep learning

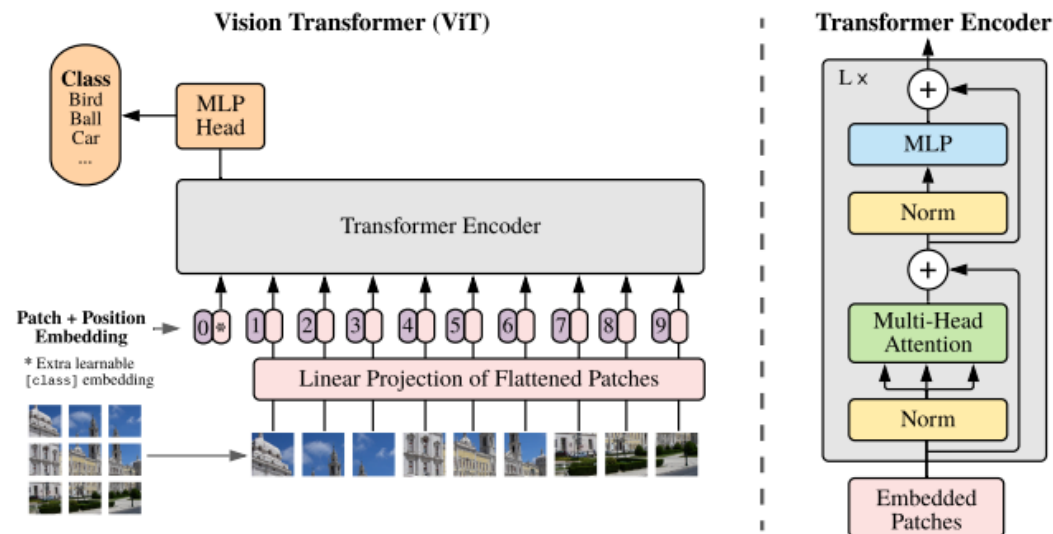
- 2018: Breakthroughs in deep learning for natural language processing (sequences)
 - Transformer architectures and pre-training -> fine-tuning.
 - State-of-the-art on 11 NLP benchmarks.



Devlin et al. 2018. BERT.

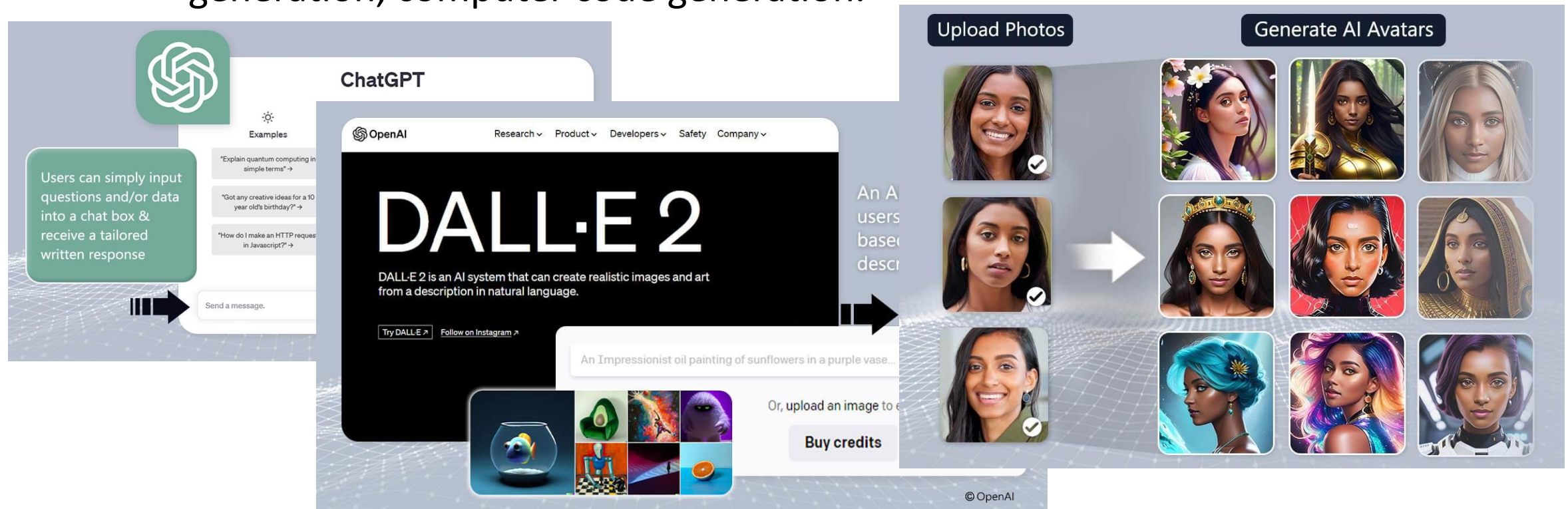
Breakthroughs in deep learning

- 2021: Vision Transformer (ViT)
- Transformer applied to image classification task.
- Various of transformer such as SwinTransformer, ViTAE Transformer has been introduced for computer vision.



Breakthroughs in deep learning

- 2022: AI-Generated Content
 - AI-Generated Content refers to content generated by artificial intelligence
 - Text generation, image generation from natural language, Audio/Video generation, computer code generation.



Breakthroughs in deep learning

- 2025: GPT-4o Image Generation
 - Combining joint image-text modeling with aggressive post-training for strong visual fluency and context-aware generation.

A wide image taken with a phone of a glass whiteboard, in a room overlooking the Bay Bridge. The field of view shows a woman writing, sporting a tshirt with a large OpenAI logo. The handwriting looks natural and a bit messy, and we see the photographer's reflection.

Read more



Best of 8



A candid paparazzi-style photo of Karl Marx hurriedly walking through the parking lot of the Mall of America,...

Breakthroughs in deep learning

- 2025: Toward World Simulation
 - Sora 2: Advancing video generation toward stronger world simulation through large-scale pre-training and post-training on video data.



Prompt: an astronaut golden retriever named Sora levitates around an intergalactic pup-themed space station with a tiny jet back that propels him. gorgeous specular lighting and comets fly through the sky, retro-future astro-themed music plays in the background. light glimmers off the dog's eyes. the dog initially propels towards the space station with the doors opening to let him in. the shot then changes. now inside the space station, many tennis balls are flying around in zero gravity. the dog's astronaut helmet opens up so he can grab one. 35mm film, the intricate details and texturing of the dog's hair are clearly visible and the light of the comets shimmers off the fur.



Prompt: a man does a backflip on a paddleboard

Breakthroughs in deep learning

- 2025: General-Purpose Humanoid Robots
 - Humanoid robots are evolving from task-specific machines into general-purpose assistants.
 - They can understand natural-language instructions and perform diverse tasks such as object manipulation, material handling, packaging, and household assistance.



Humanoid robots performing coordinated manipulation tasks in an industrial environment.



A humanoid robot recognizes, grasps, and places a household object.

Fundamentals of deep learning

- Deep learning is a hype or reality?



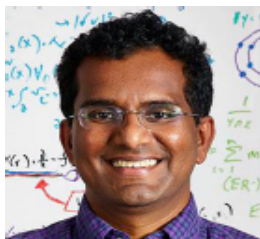
I have worked all my life in Machine Learning, and I've never seen one algorithm knock over benchmarks like Deep Learning

– Andrew Ng (Stanford & Baidu)



Deep Learning is an algorithm which has no theoretical limitations of what it can learn; the more data you give and the more computational time you provide, the better it is

– Geoffrey Hinton (Google)



Human-level artificial intelligence has the potential to help humanity thrive more than any invention that has come before it

– Dileep George (Co-Founder Vicarious)

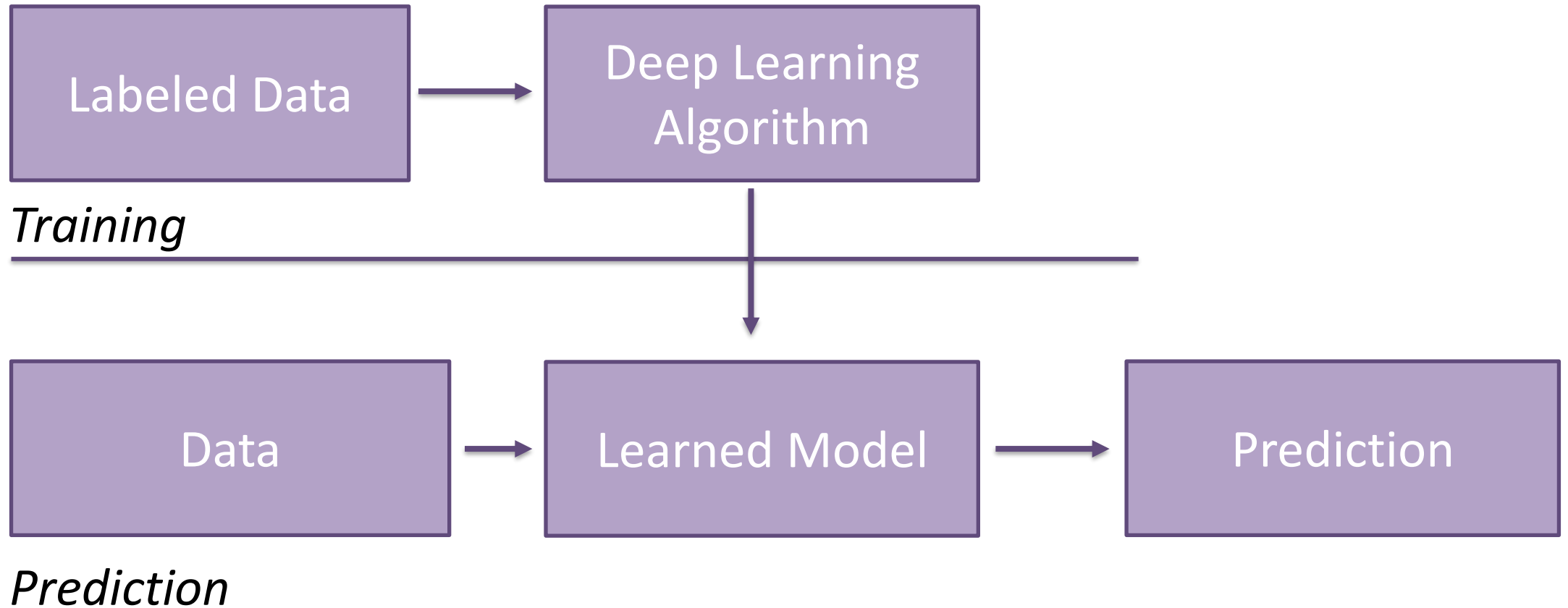


For a very long time it will be a complementary tool that human scientists and human experts can use to help them with the things that humans are not naturally good at

– Demis Hassabis (Co-Founder DeepMind)

Fundamentals of deep learning

- Deep learning – introduction

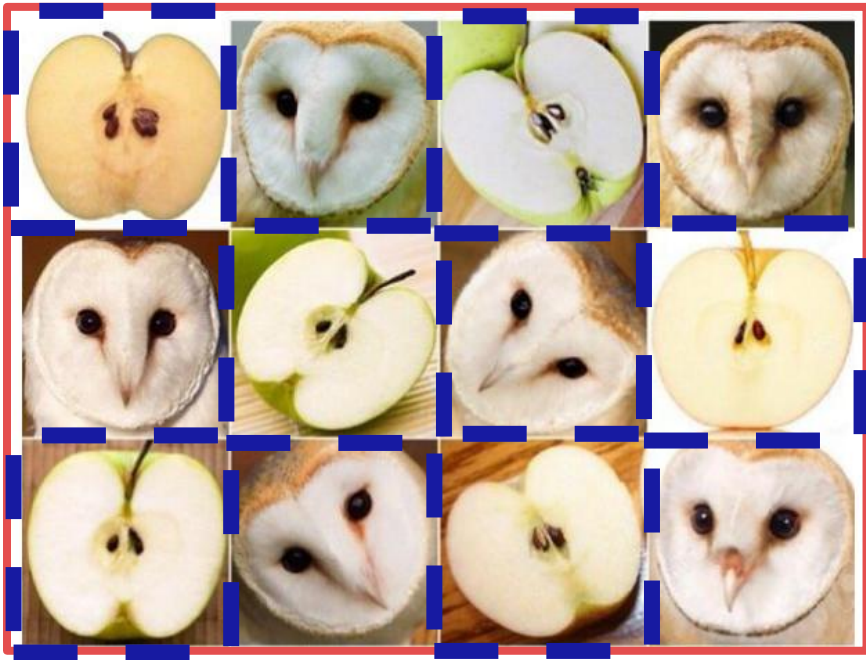


Provides **various techniques** that can learn from and make predictions on data

Fundamentals of deep learning

- Deep learning – Classification task

Apple



Labeled Data

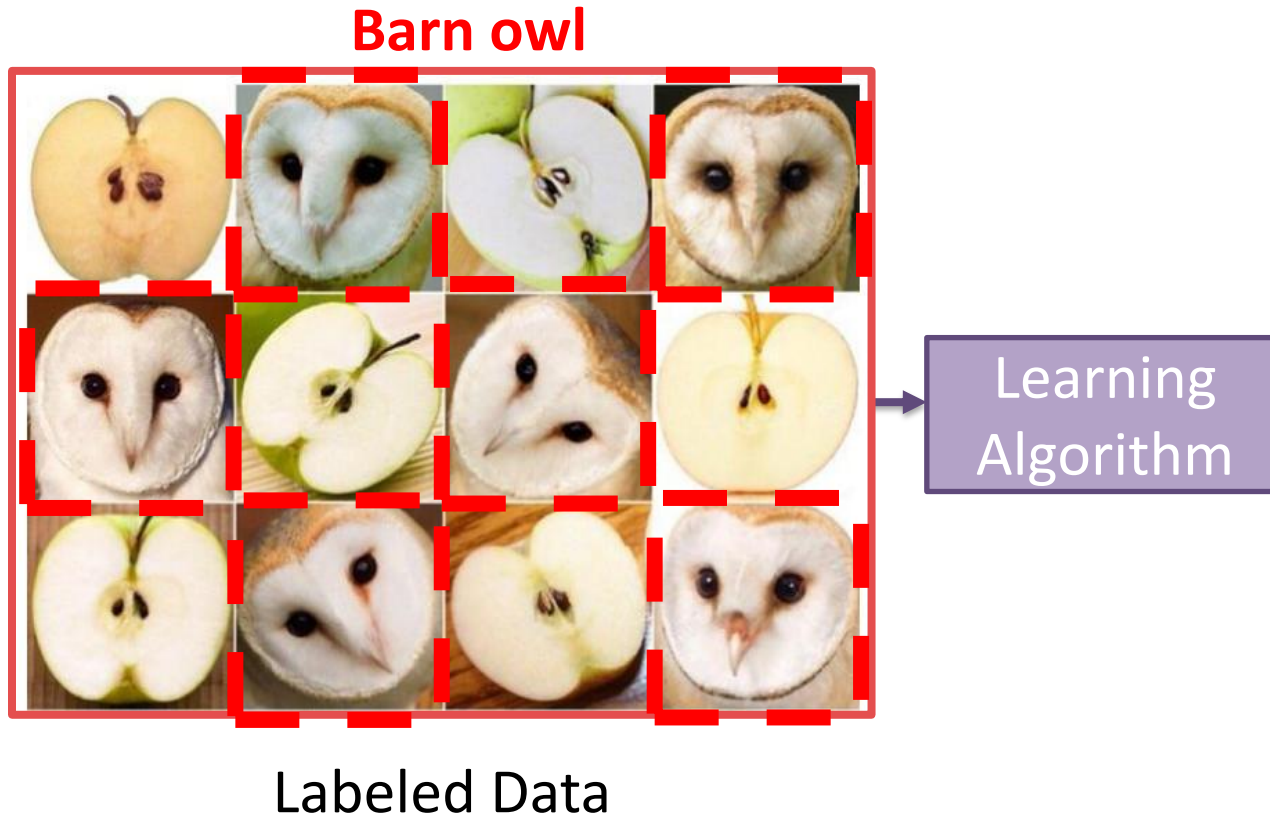
Learning
Algorithm

Training

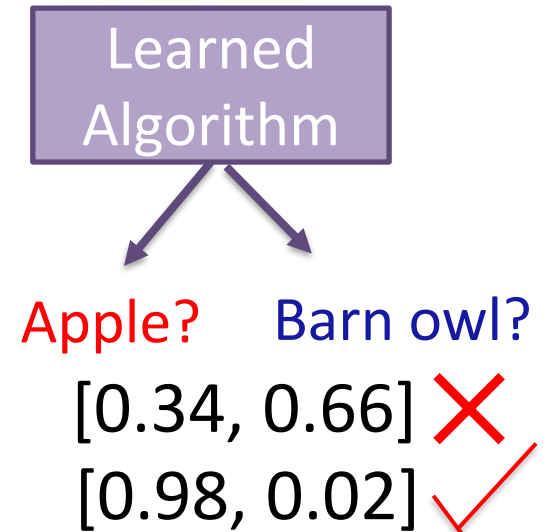
Prediction

Fundamentals of deep learning

- Deep learning – Classification task



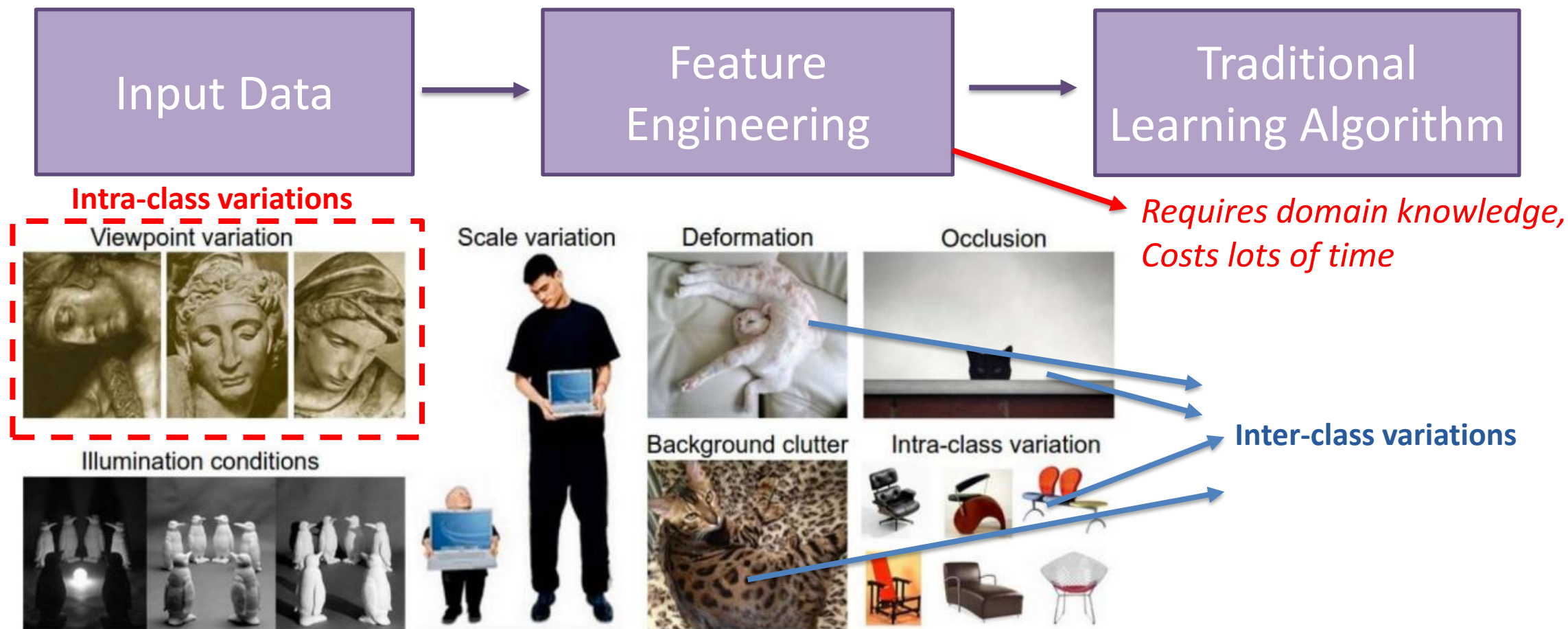
Training



Prediction

Fundamentals of deep learning

- Why deep learning? – data driven, no more feature engineering



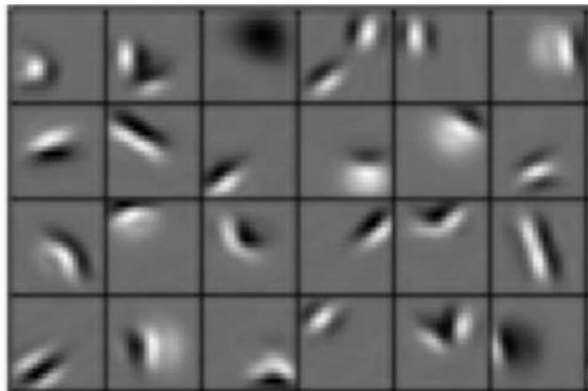
Fundamentals of deep learning

- Why deep learning? – data driven, no more feature engineering

Can we learn *a hierarchy of features* directly from the data?



Low level features



Edges, dark spots

Mid level features



Eyes, ears, nose

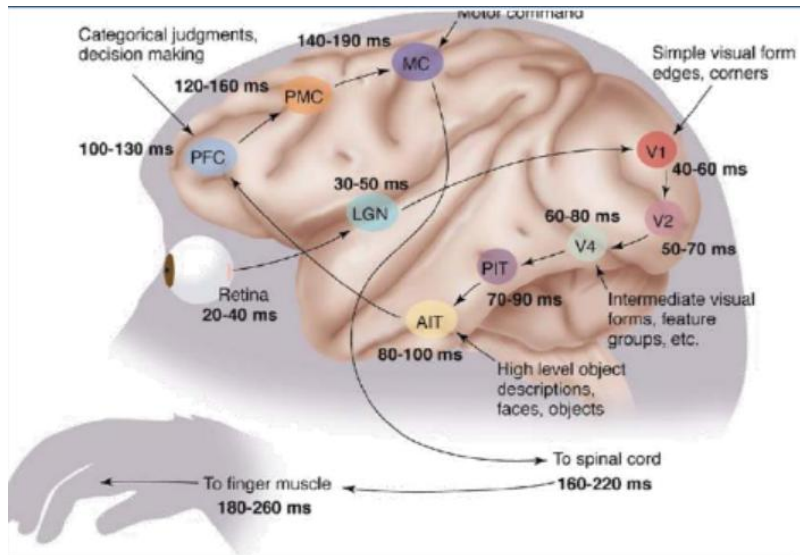
High level features



Facial structure

Fundamentals of deep learning

- Deep learning – Inspired by the brain

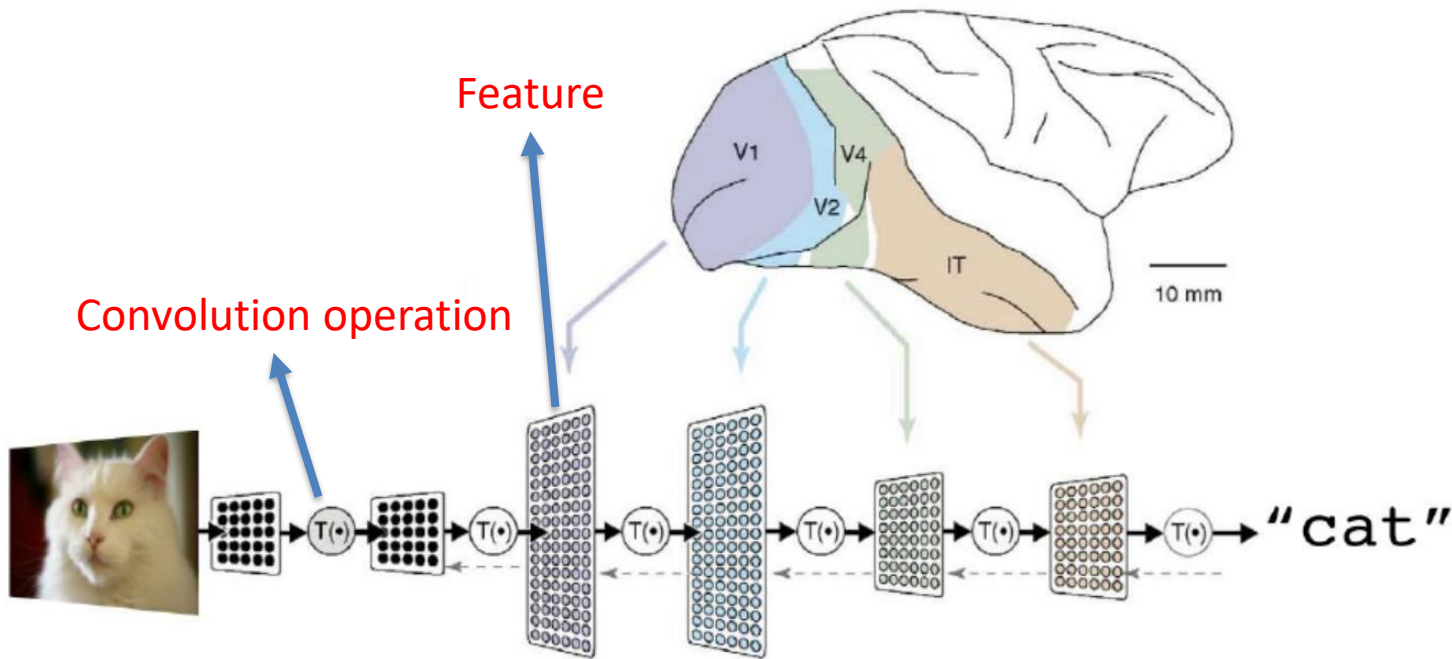


- The **first hierarchy of neurons** that receives information in the visual cortex are sensitive to specific edges.

- Our brain has lots of neurons connected together and the **strength of the connections** between neurons represents **long term knowledge**.

Fundamentals of deep learning

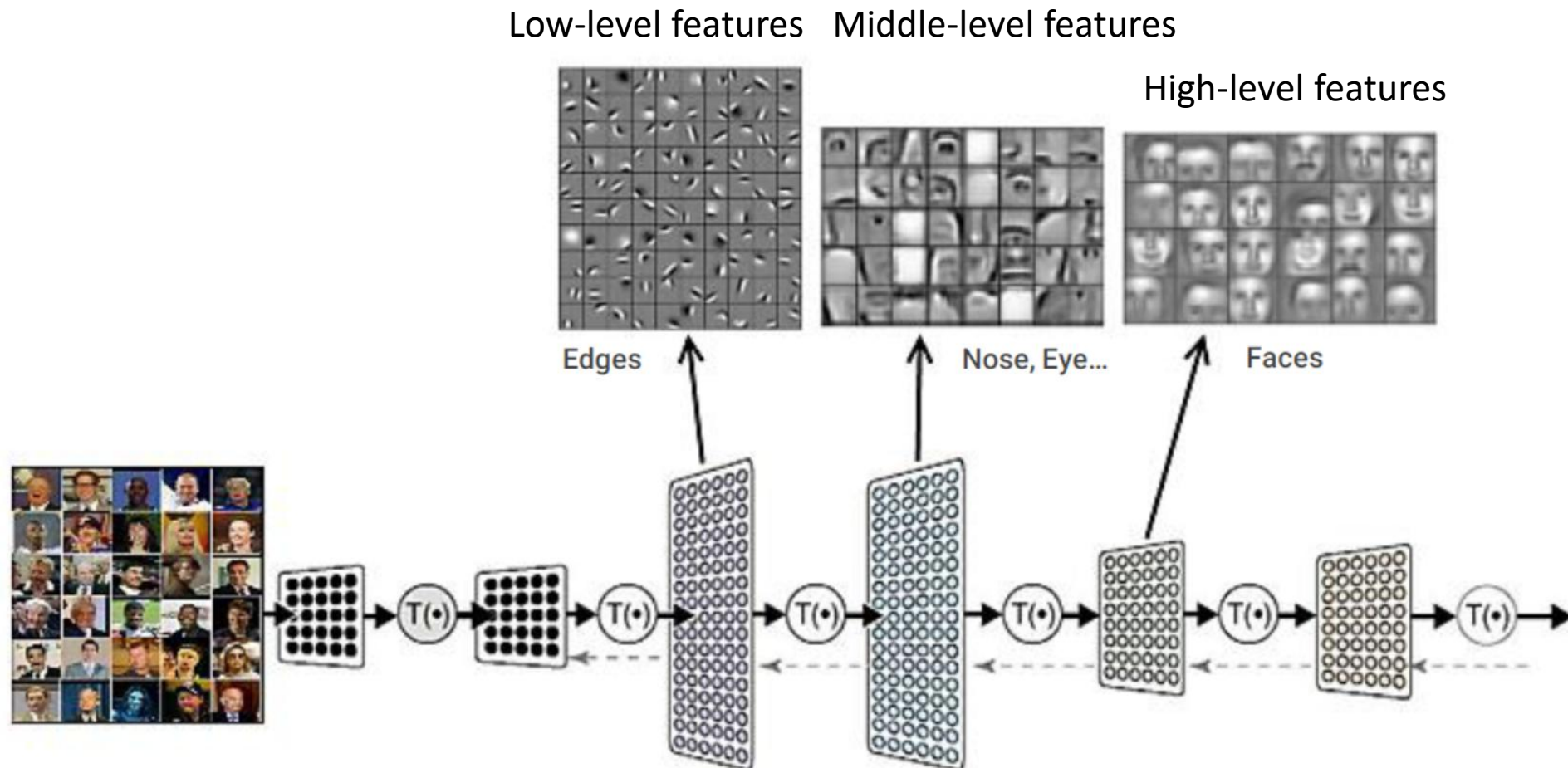
- Deep learning – Inspired by brain



A deep neural network consists of a **hierarchy of layers**, whereby each layer **transforms the input data** into **more abstract features** (e.g. edge -> nose -> face). The output layer make predictions based on features.

Fundamentals of deep learning

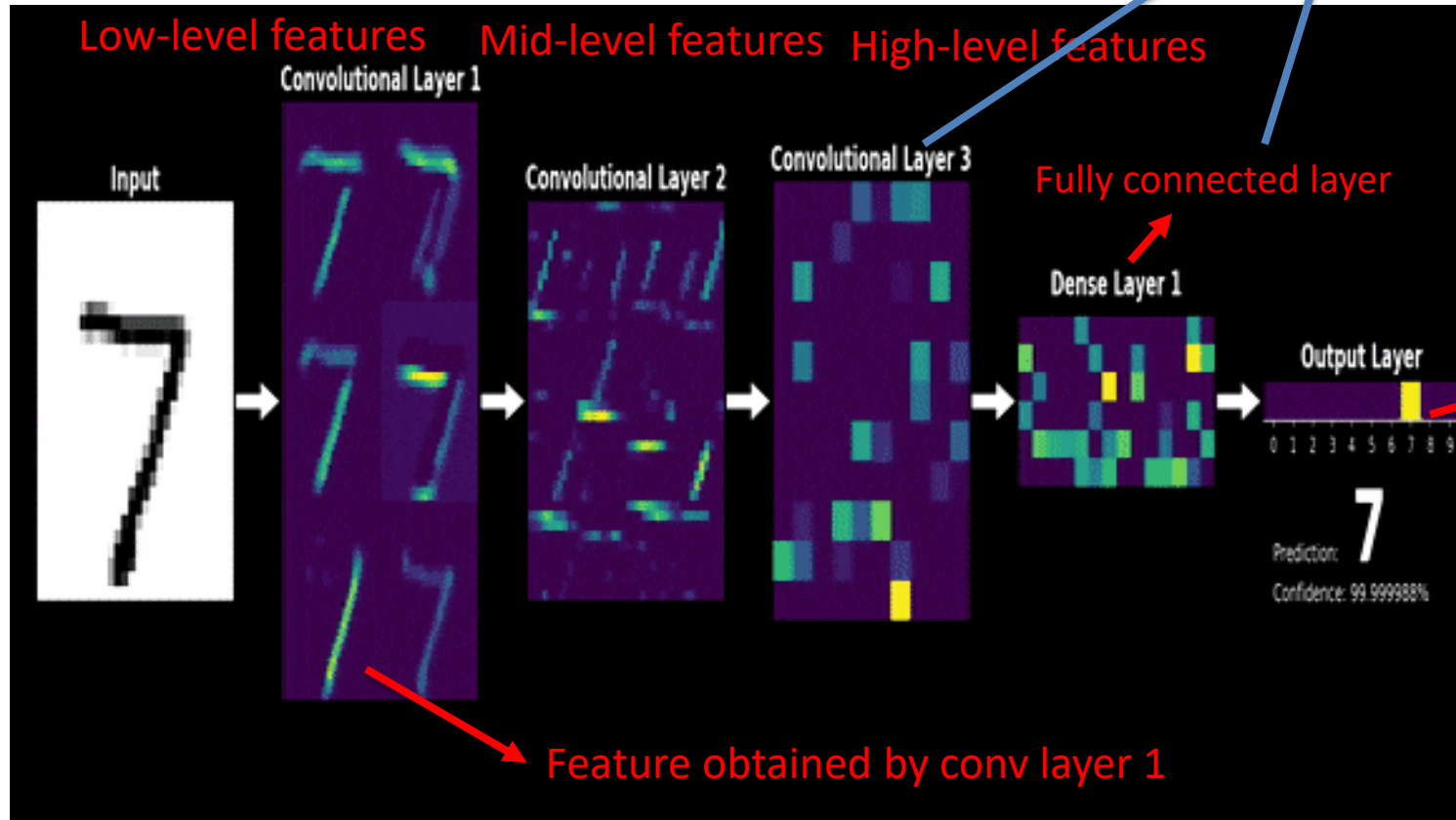
- Deep learning – what did it learn?



Fundamentals of deep learning

- Deep learning – what did it learn?

What is the differences?



Showing the probabilities that the input belongs to each class.

A vector

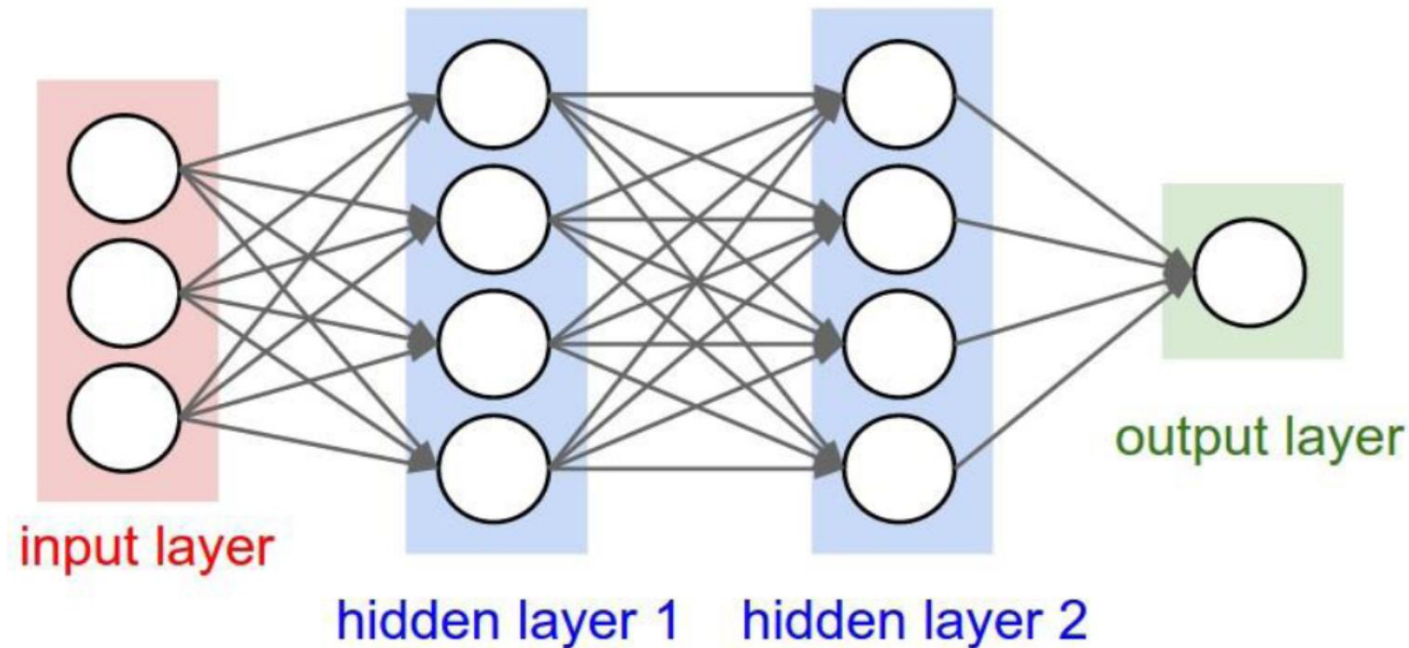
Input: 1

Output: [0.0, 0.8, 0.1, 0.0, 0.1, 0.0, 0.0, 0.0, 0.0, 0.0]

Output has 0.8 probability to be class 1.
The summation of these probabilities should equal to 1.

Fundamentals of deep learning

- Deep learning – an artificial neural network



Fully-connected layers have many parameters and difficult to optimize.

- Consists of **one input**, **one output** and **multiple fully-connected layers**.
- Each layer is represented as **a series of neurons** and **progressively extracts higher and higher-level features** of the input until the final layer essentially makes a decision about what the input shows.
- The **more layers** the network has, the **higher-level features** it will learn.

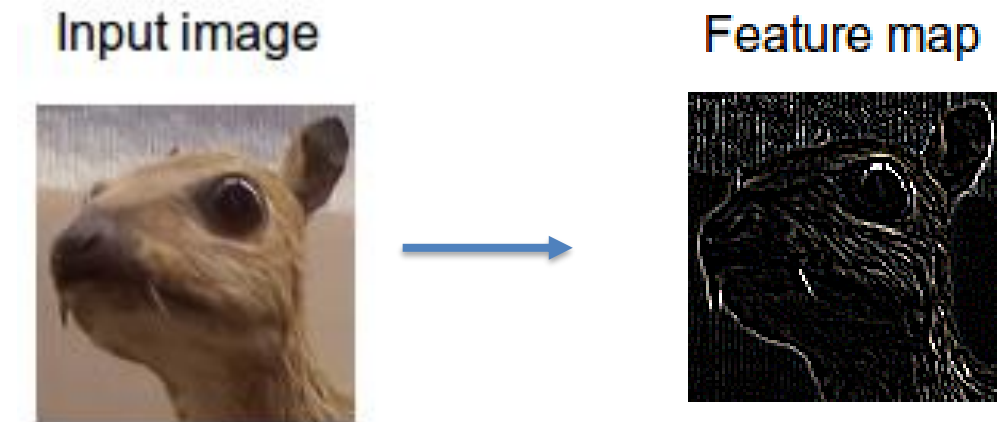
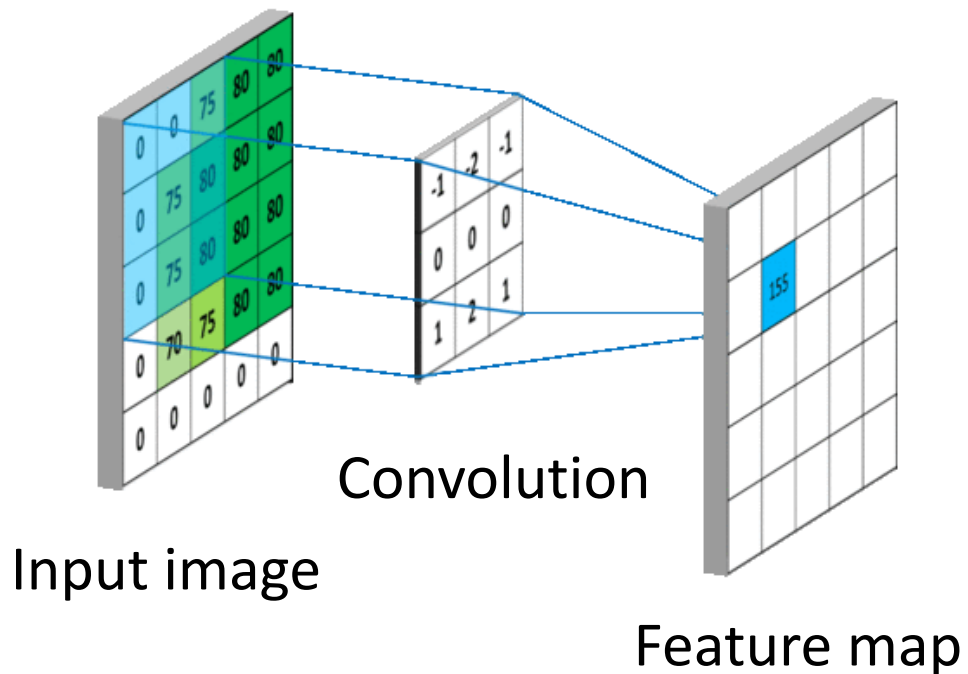
Fundamentals of deep learning

- Convolutional neural network (CNN) – Convolution layer

- The **most widely used** artificial neural network for imaging tasks.
- Takes the spatial relationship of images.
- Fewer parameters** than fully connected layer.

Aim:

Conv is used as **feature extractor**.
To **filter important and high-level** information.



Fundamentals of deep learning

- Convolutional neural network (CNN) – Convolution layer

Convolution operation

$\times 1$	$\times 0$	$\times 1$
$\times 0$	$\times 1$	$\times 0$
$\times 1$	$\times 0$	$\times 1$

Filter



convolution kernels (*network weights*)

1 _{$\times 1$}	1 _{$\times 0$}	1 _{$\times 1$}	0	0
0 _{$\times 0$}	1 _{$\times 1$}	1 _{$\times 0$}	1	0
0 _{$\times 1$}	0 _{$\times 0$}	1 _{$\times 1$}	1	1
0	0	1	1	0
0	1	1	0	0

Image

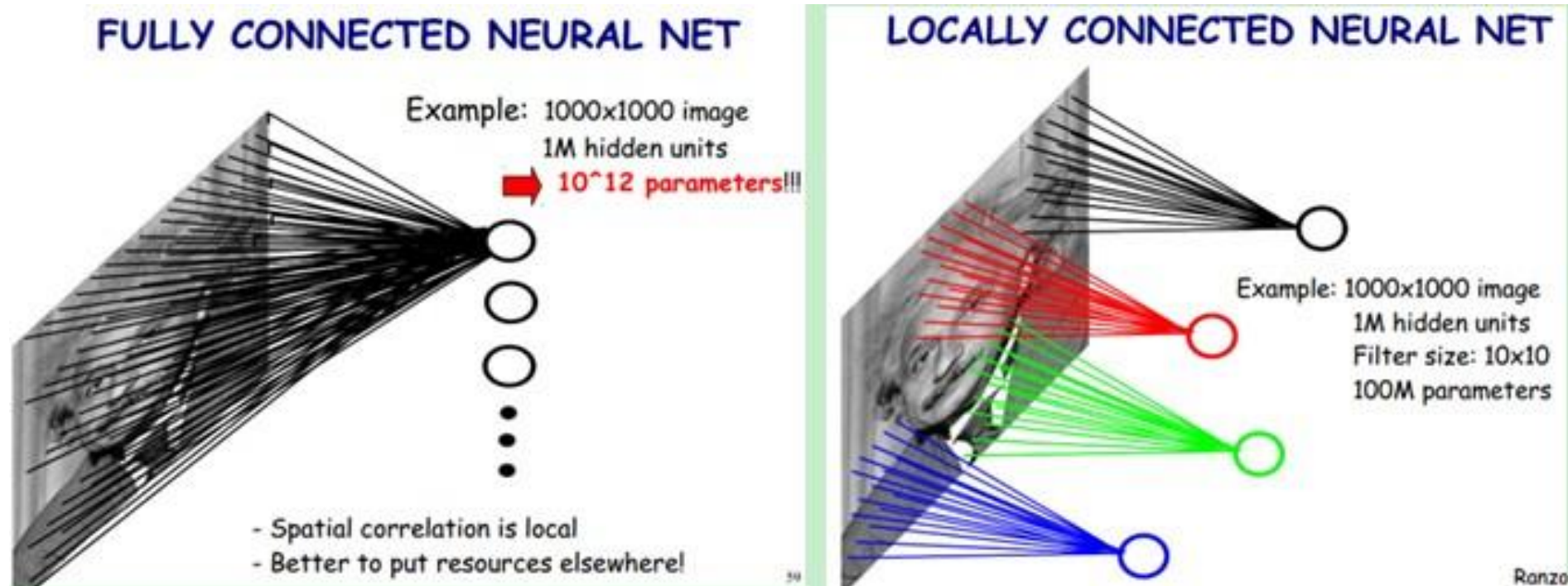
4		

Convolved
Feature

Fundamentals of deep learning

- Convolutional neural network (CNN) – Fully connected layer

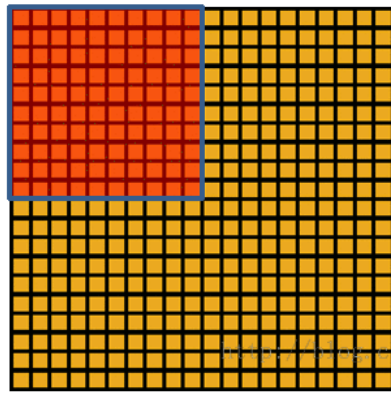
Convolution operation



Fundamentals of deep learning

- Convolutional neural network (CNN) – Pooling layer

Pooling operation



Convolved
feature

1	

Pooled
feature

Max pooling

Average pooling

Input

7	3	5	2
8	7	1	6
4	9	3	9
0	8	4	5

maxpool

Output

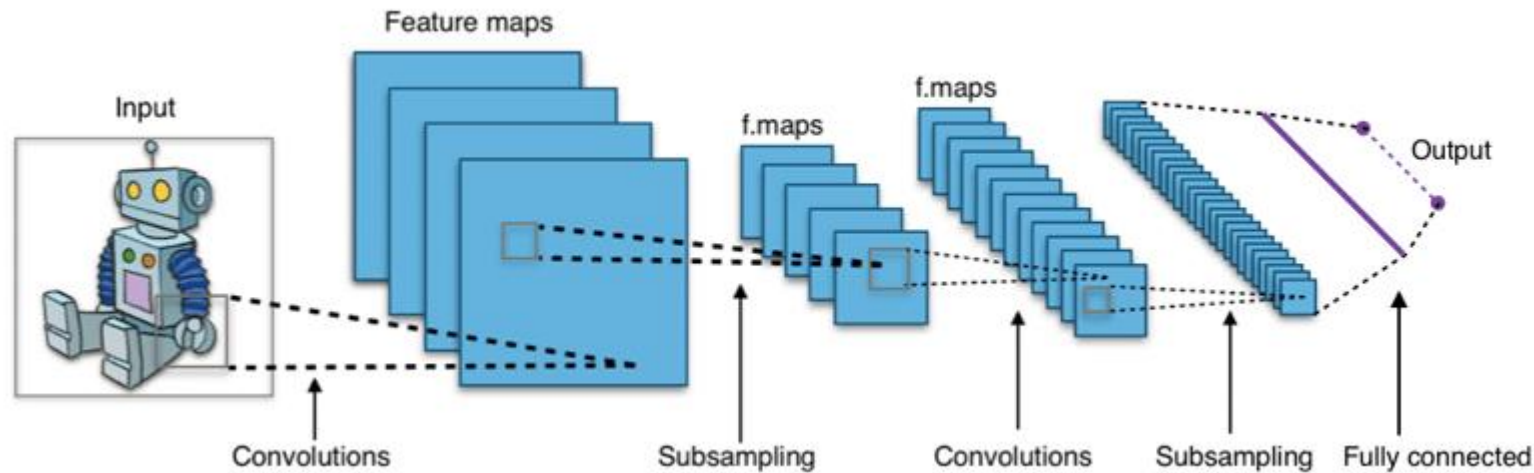
8	6
9	9

Aim:
reduce the parameters

Fundamentals of deep learning

- Convolutional neural network (CNN)

A simple example for image classification

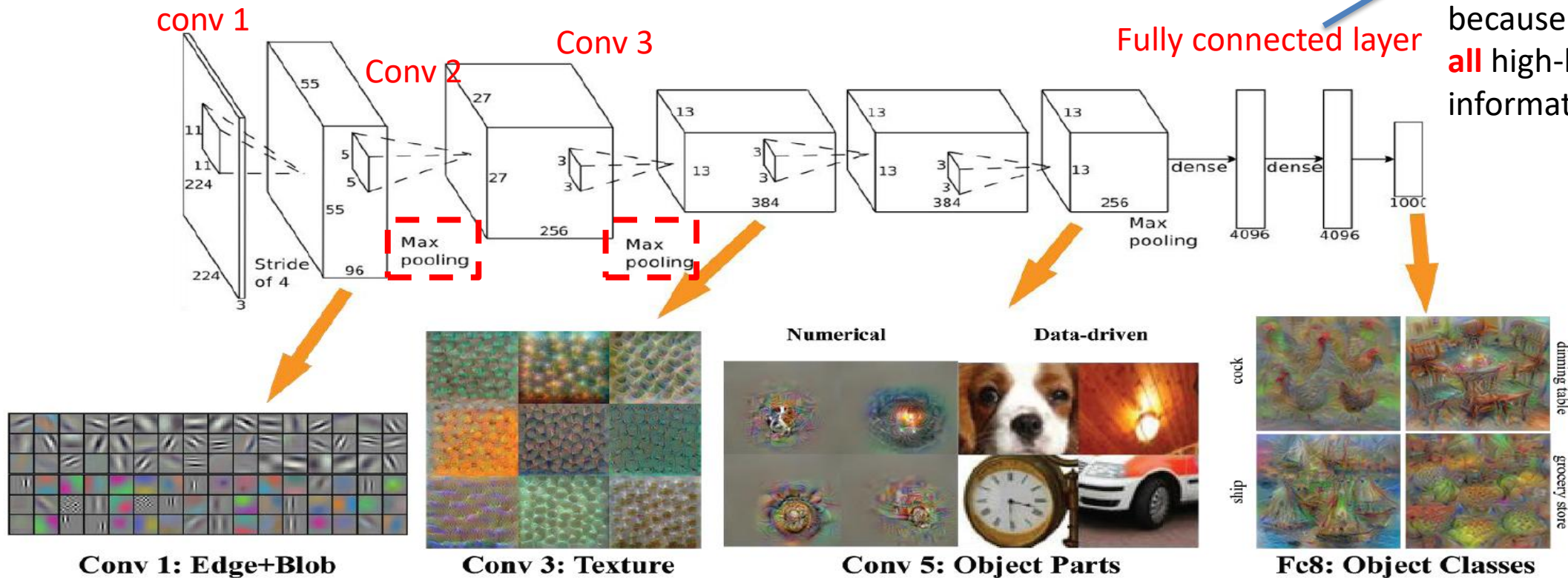


- Consists of **convolution layers**, **max-pooling layers**, **fully connected layer**.
- The **input is the images** and **output is the vector**, indicating the probabilities belong to each class.

Fundamentals of deep learning

- Convolutional neural network (CNN) – AlexNet

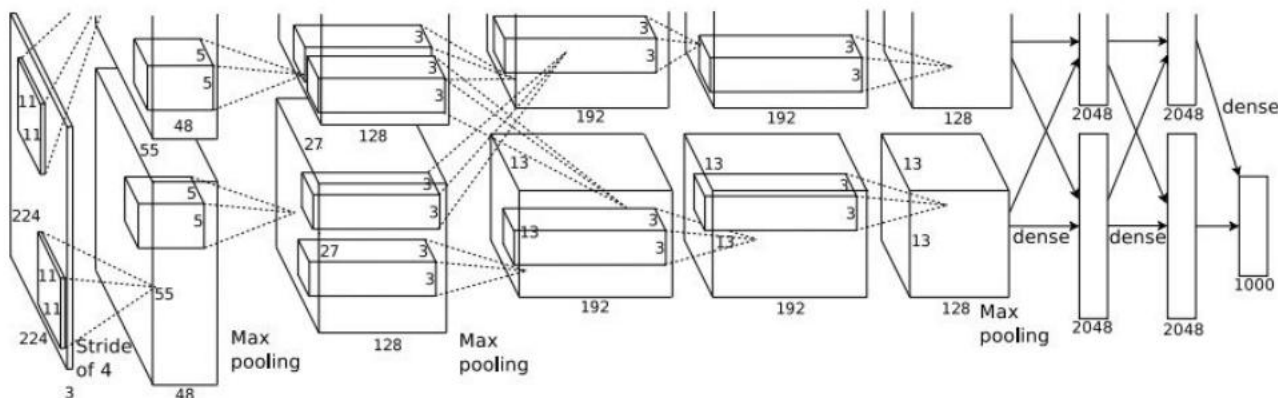
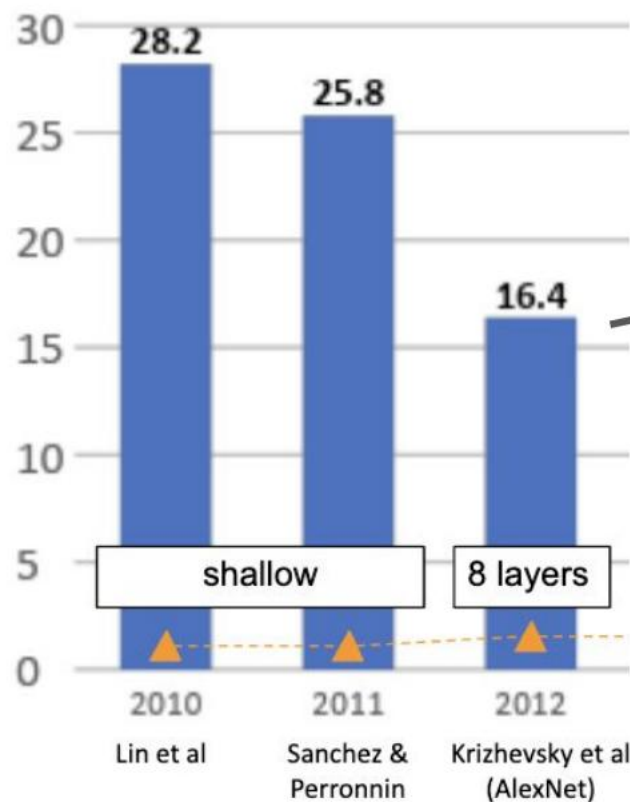
Always used as the final layer because we need **all** high-level information.



- AlexNet was proposed in 2012
- Achieved very high accuracy on ImageNet challenge.

Fundamentals of deep learning

- Convolutional neural network (CNN) – AlexNet

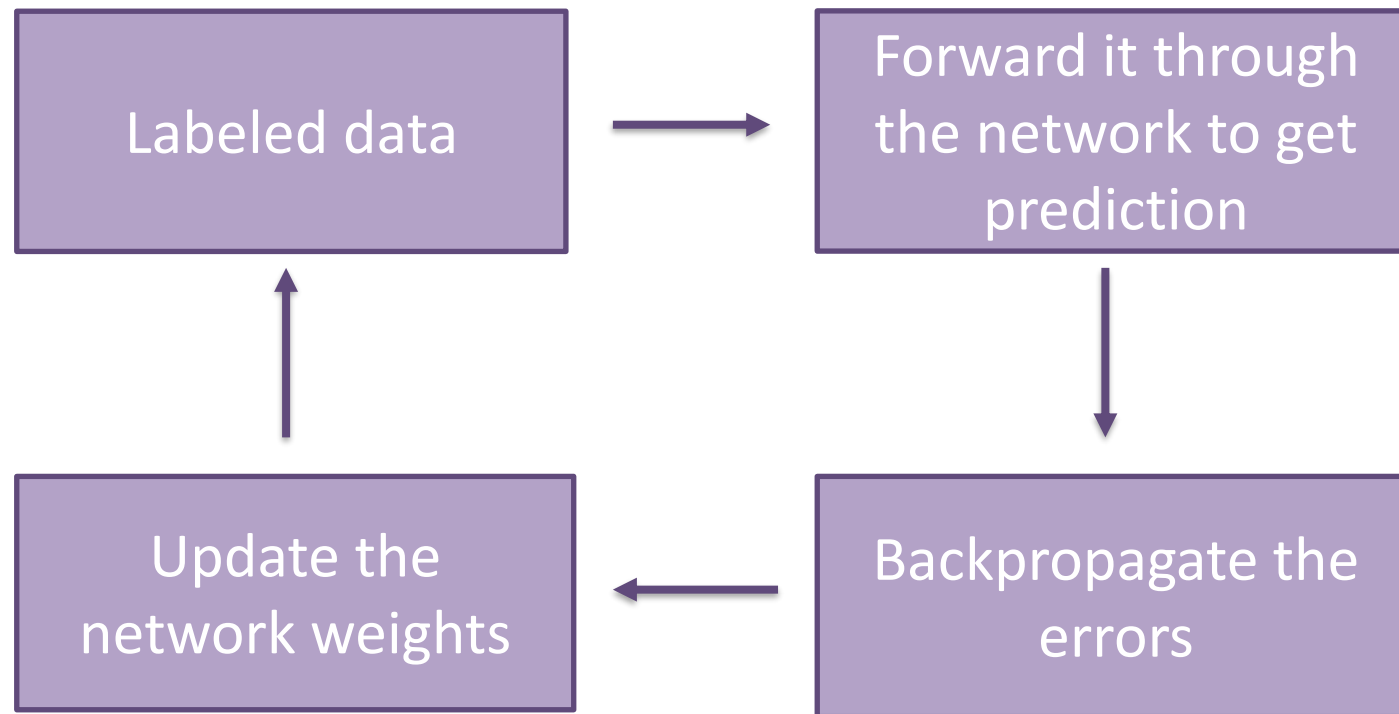


Krizhevsky et al. 2012. 8-layer “AlexNet”.

- ImageNet Visual Recognition Challenge results.

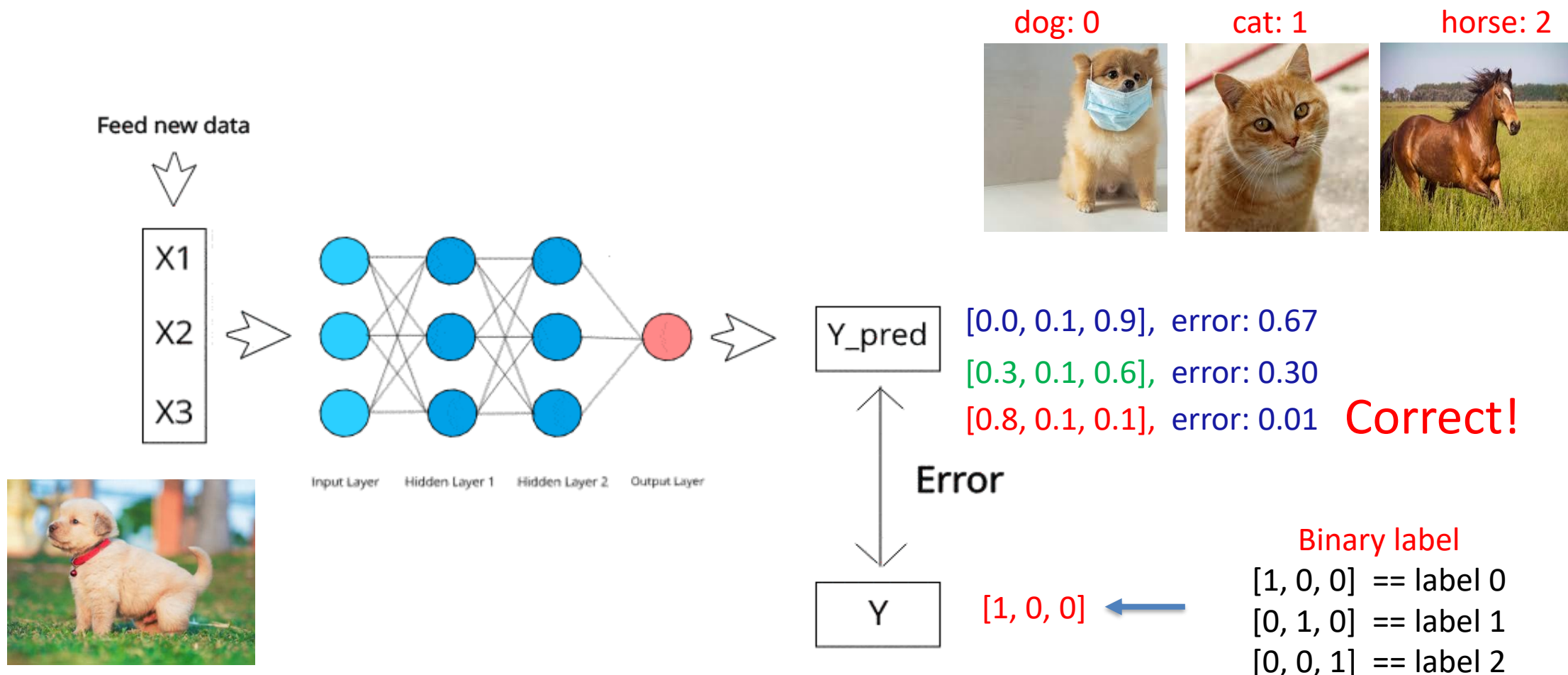
Fundamentals of deep learning

- Convolutional neural network (CNN) – how to train



Fundamentals of deep learning

- Convolutional neural network (CNN) – how to train?



A week in the life of ELEC 5690

Tuesday

Lecture
(09:00 - 11:50 AM)

Room: Rm 2610 (Lift 31-32)

Optional Textbook



- [1] Deep Learning by Goodfellow, Bengio, and Courville. <https://www.deeplearningbook.org/>
- [2] Medical imaging signals and systems, Jerry L. Prince and Jonathan M. Links, 2015
- [3] Saxena, Ankur, and Shivani Chandra. Artificial intelligence and machine learning in healthcare. Springer Singapore, 2021.
- [4] Xiao, Cao, and Jimeng Sun. Introduction to Deep Learning for Healthcare. Springer Nature, 2021.
- [5] Zhou, Kevin, Hayit Greenspan, and Dinggang Shen, eds. Deep learning for medical image analysis. Academic Press, 2017.
- [6] Lee, Gobert, and Hiroshi Fujita, eds. Deep learning in medical image analysis: challenges and applications. Vol. 1213. Springer Nature, 2020.

Assessment

Programming Assignment (20%)

- You are required to finish the **coding** of two tasks related to medical image **classification** and **segmentation**, and finish a summary report, including quantitative and qualitative results
- Provisional Duration: Sep. 20th ~ Oct. 16th
- Required knowledge: Basic knowledge about deep learning, medical image processing; familiar with Python, **PyTorch**, related python libraries, e.g., Numpy, OpenCV-python, SimpleITK

Paper Reading & Sharing (20%)

- You will be assigned with one paper related to medical image analysis, You are required to read it and make a 15-min oral presentation (10-min presentation + 5-min Q&A) in class, the paper sharing is planned to start on Sep. 22nd (4th class)
- Provisional date on paper list release: Sep. 10th
- Required knowledge: Basic knowledge about deep learning network, medical image analysis, research survey ability

Midterm Exam (30%)

- Fundamentals of deep learning and medical image analysis covered in class
- Closed-book exam with one A4 sheet of notes allowed

Assessment

Final Project & Presentation (25%)

- This is a group project (1-2 students per group), each group is required to choose one topic related to medical image analysis. For each project, students need to survey related works, run a baseline model on the chosen datasets, propose a **new** method/architecture/strategy, write a summary **report**, give a face-to-face **presentation**
- Provisional presentation : 2nd week of Dec.
- Required knowledge: Advanced understanding of deep learning network and medical image analysis, research survey ability, familiar with Python, **PyTorch**, related python libraries, e.g., Numpy, OpenCV-python, SimpleITK

Class Attendance (5%)

Late Policy

- 3 free late days to use on assignments
- Once free late days are exhausted, 25% penalty per day

Opportunities

Top conferences for medical image processing

MICCAI: The Medical Image Computing and Computer Assisted Intervention Society

<https://miccai.org/index.php/events/upcoming-conferences/>

ISBI: International Symposium on Biomedical Imaging

<https://biomedicalimaging.org/2027/>

IPMI: International Conference on Information Processing in Medical Imaging

<https://2027.ipmi-conf.org/>

Besides, you can also publish in more generic computer vision conferences

CVPR: IEEE/CVF Conference on Computer Vision and Pattern Recognition

<https://cvpr.thecvf.com/Conferences/2026>

ICCV: International Conference on Computer Vision

<https://iccv2025.thecvf.com/>

NeurIPS: Conference on Neural Information Processing Systems

<https://neurips.cc/>

Opportunities

Besides, you can also publish in more generic computer vision conferences

IJCAI: International Joint Conference on Artificial Intelligence

ECCV: European Conference on Computer Vision

AAAI: Association for the Advancement of Artificial Intelligence

Journals:

- **IEEE Transactions on Pattern Analysis and Machine Intelligence**
- **International Journal of Computer Vision**
- **IEEE Transactions on Medical Imaging**
- **Medical Image Analysis**
- **Nature Biomedical Engineering**
- **Nature Machine Intelligence**
- **Nature Communications**
- **Journal of Biomedical and Health Informatics**
- ...

Always be Honest and act with Integrity

Academic Integrity 學術誠信

The HKUST Academic Honor Code

Honesty and integrity are central to the academic work of HKUST. Students of the University must observe and uphold the highest standards of academic integrity and honesty in all the work they do throughout their program of study.

◆

As members of the University community, students have the responsibility to help maintain the academic reputation of HKUST in its academic endeavors.

◆

Sanctions will be imposed on students, if they are found to have violated the regulations governing academic integrity and honesty.

Over the long term being honest and decent will be the most rewarding thing you can ever do!!

Questions are more important than answers

Questions are more important than answers

Knowledge and answers, by themselves, are no longer enough

Curiosity is essential

Asking probing questions is essential

Knowing how to find the answers to your questions is critical

Collaboration Policy

- Rule 1: **Don't look at** solutions or code that are not your own; everything you submit should be your own work
- Rule 2: **Don't share** your solution code with others; however discussing ideas or general strategies is fine and encouraged
- Rule 3: Indicate in your submissions anyone you worked with
- Turning in something late / incomplete is better than violating the honor code

Course Philosophy

- Thorough and Detailed
 - Understand how to write from scratch, debug, and train convolutional and other types of deep neural networks
 - We prefer to write from scratch, rather than rely on existing implementation.
- Practical
 - Focus on practical techniques for training and debugging neural networks
 - Will use state-of-the-art software tools like Pytorch or Tensorflow
- State of the art
 - Most material we cover is research published in the last 5 years.

Course Structure

- **Fundamentals of Deep Learning**
 - Details of how to implement and train different types of networks
 - Fully-connected networks, convolutional networks, recurrent networks
 - How to train and debug, very detailed
- **Fundamentals of Medical Images and Healthcare**
 - Introduction of different type of medical images
 - Disease screening and segmentation from medical images
 - An overview of healthcare applications
- **Applications and “Researchy” topics**
 - Disease screening, image segmentation, surgical videos, etc.
 - Annotation-efficient deep learning for medical image analysis
 - Self-attention and generative models

Next time:
Deep Learning Fundamentals